

BLOCK 2
ENVIRONMENT AND MOVEMENTS

BLOCK 2 ENVIRONMENT AND MOVEMENTS

Introduction

Unit 6 International Environmental Policies, Agreements and Treaties: Deals with International Environmental Policies, Agreements and Treaties. This unit comprehends the need for International Environmental Law; make you to understand the implications of Stockholm Conference, Rio Conference and Johannesburg Treaty. It highlighted the Stockholm Conference 26 principles, includes fundamental right to freedom, international equity, maintenance, restoration and improvement of renewable resources. Agenda 21 also comprehended as a part of Rio declaration. The unit concluded by explaining some important international environmental law agreement like CITES, Cartagena Protocol on bios safety, Montreal Protocol and Rotterdam Convention.

Unit 7 Environmental Sustainability deals with the concept of sustainable development, steps to achieve sustainable development through specific targets and indicators, the international efforts towards a sustainable future and different ethical viewpoints towards the environment.

Unit 8 Environmentalism: Deals with Environmentalism. The unit details on the concepts in environmentalism and the people involved in the environmental movements. The unit also discusses on environmental inequality and the factors responsible for the same. The ecocentric, biocentric and anthropocentric views have been discussed for better understanding. Finally, the unit also touched upon deep ecology, social ecology, radical ecology.

Unit 9 Social and Civil Society Movements in India: Discussed about Social and Civil Society Movement in India. It introduces you the social and civil society movements in environment. It pronounced The Bishnoi Movement as one of the earliest Environmental movement which concerned for Environmental Protection, ensures improvement in the living conditions of the desert community in Rajasthan. How Colonial India witnessed forest policies, tribal land displacement and deprivation of livelihood of Adivasi was depicted. It elaborated the environmental movements of contemporary India like Chipko Movement, Narmada Bachao Andolan movements to protest against destruction of natural resources conservation.

UNIT 6 INTERNATIONAL ENVIRONMENTAL POLICIES, AGREEMENTS AND TREATIES

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6.0 INTRODUCTION

Global environmental issues are examined under the subject of International Environmental law. Debates over environmental issues are based on the principles of international law, numerous international agreements and declarations. International Environmental law is not a separate branch of law but a part of the International law which governs relations between states. Environmental international law has evolved from a number of sources. The most important source is the customary international law. Customary international laws are the norms followed by most countries as a matter of custom. They are responsible for the common thread that binds all countries.

Other sources of the international law include general principles of law recognized by civilized nations. International law is also evolved from subsidiary sources such as decisions of courts and tribunals and writing of jurists.

Numerous legally binding international agreements cover a wide range of environmental issues. They cover different aspects like: terrestrial, marine, atmosphere to biodiversity protection. Most of the International environmental agreements are generally multilateral. Protocols are subsidiary agreements that are based on the primary treaty. Protocols are available in various subsets of international law. These find practical applicabilities in the issues relating to environment. They enable the incorporation of recent scientific advances. Kyoto Protocol is one of the most popular protocols in international environmental law. United Nations Framework Convention on Climate Change follows the Kyoto protocol.

Some important conferences like the United Nations Conference on the Human Environment (1972), United Nations Conference on Environment and Development (1992); and World Summit on Sustainable Development (2002) have had tremendous impacts. International organizations with well defined roles are also created due to multilateral environmental agreements. These bodies are responsible for implementing the agreement. Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) and the International Union for Conservation of Nature (IUCN) are the products of such agreements.

The opinions of international courts and tribunals provide the background for International environmental law. The declarations, codes, guidelines are considered as soft law. They are not binding. Soft laws result because of the following reasons: (1) lack of consensus in arriving at a universal law, (2) lack of one supreme body for making law, and (3) diverse cultures, religion etc. However, the soft laws facilitate collective action by states when required, without restricting their freedom of action.

6.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- comprehend the need for international environmental law;
- understand the implications of Stockholm conference, Rio conference and Johannesburg treaty; and
- explain some other important international environmental legislations.

6.2 STOCKHOLM CONFERENCE

The idea of holding a United Nations conference centred on human interactions with the environment is attributed to the proposal of Sweden in the United Nations Economic and Social Council in the year 1968. After the idea got support from the ECOSOC, the General Assembly decided in 1969 to hold the conference in 1972. The General Assembly proposed that the conference should result in “stimulating and providing guidelines for action by national government and international organizations”. Extensive background work by 115 governments for the conference lasted for about four years. Stockholm, Sweden hosted the United Nations Conference on the Human Environment in June 1972. It was the first comprehensive conference on international environmental issues with the backing of the UN. This event

marked a major milestone in international environmental politics.

6.2.1 Positive Outcomes

The conference agreed upon the following:

1. A Declaration about environment and development (26 principles);
2. An clear cut plan of action (109 recommendations), and
3. Resolutions relating to the financial and Institutional arrangements.
 - a. The Stockholm resolution supported the pre-Stockholm treaties and conventions (relating to the marine pollution, transboundary air pollution, endangered species).
 - b. Action plan dealing with creation of institutions and co-operation amongst already existing ones. It also gave an action plan for the international community.
 - c. The conference gave 26 principles, which include:
 - d. Fundamental right to freedom, equality and adequate conditions of life, in an environment of a quality that permits a life of dignity and well-being.
 - e. Intergenerational equity, maintenance, restoration and improvement of renewable resources.
 - f. Appeal to the people to manage the environment, so as to avoid irreversible damage to the ecosystem.
 - g. Recognition of the relationship between economic, social and environmental development. Encouraging development through transfer of technology and financial aid.
 - h. That the States should cooperate in developing international environmental law.
 - i. The States should adopt and implement suitable environmental standards.
 - j. The International Organizations should co-ordinate in such activities that facilitate better environmental management

The UN General Assembly considered the report submitted by the Stockholm Conference. The recommendations were considered in the development of environment related standards and activities at all levels: regional, national and global level. It led to the development of a precursor to the formation of an International Environmental Organization through the United Nations Environment Programme. This directed the future course of the environmental policies in Europe. However, the Stockholm conference did not result in the adoption of binding legal obligations for the State parties.

6.3 RIO CONFERENCE

The Rio conference was held 20 years after the first International conference on environmental issues at Stockholm. The main objective of the Rio conference was to stop environmental degradation of the planet. Following

the Stockholm conference, UN in the year 1983 set up a Commission on Environmental and Development led by Brundtland from Norway. The Commission suggested the concept of Sustainable Development. This is defined as “the development which meets the needs of the present generation without compromising the ability of the future generations to meet their own needs”. Based on the Brundtland report, 1987- the UN General Assembly called for the convention of UNCED (United Nations Conference on Environment and Development). UNCED was held at Rio De Janeiro, Brazil in June, 1992 represented by 172 governments. Several representatives of non-governmental organizations (NGOs) attended. It focussed on balancing development that support socio-economic development and avoids damage to the environment. It also stressed on the need to bring about global partnership to tackle the issues.

6.3.1 Issues Covered

The issues addressed in the Rio conference include the following:

- Comprehensive study of production patterns which include toxic components such as lead in gasoline, toxic wastes including radioactive chemicals.
- The use of other sources of energy replacing fossil fuels.
- Promoting the use of public transportation systems. This will reduce vehicular emissions, congestion in cities and the associated health risks.
- The increasing demands and diminishing water availability.

The UNCED adopted three important non-binding instruments in the forms of:

- Rio Declaration on Environment and Development
- Agenda 21
- Forest Principles

The Rio declaration on Environment and Development contains twenty seven principles. The unique feature was that it made efforts to maintain balance of priorities of the developed and developing countries. It also gave guidelines regarding principles for sustainable development. Though this is a form of soft law, the legal implications is immense as a large number of treaties, protocols, regulations and judicial decisions were based on this. Even the International Court of Justice has referred to it in a number of cases.

The adoption of Forestry principles led to the revision of the International Timber Trade Agreement in 1994. This was the first legal consensus reached on forests. It posed responsibility on the developed countries through reforestation and forest conservation. At the same time, they were given the liberty to develop the forests depending on their socio-economic needs.

6.3.2 Agenda 21

Agenda 21 is a dynamic programme and a voluntary action plan. It is called so because it is a comprehensive blue-print for local, national, regional and global

actions. The proposed actions will enable the transition to sustainable development in the 21st century. It was adopted by more than 178 Governments at the Rio Conference. It comprises of 40 chapters and hundreds of programme areas. Agenda 21 lays down the ways and means in the reduction of wasteful and inefficient consumption patterns. It recognized the need of developed countries like extending financial support and aiding technology transfer to developing countries to achieve sustainable development.

6.3.3 Important Agreements

Some legally binding agreements (Rio Convention) came into force. They are given below.

- Framework Convention on Climate Change (UNFCCC)
- Convention on Biological Diversity

One major outcome of the summit was an agreement on the Climate Change Convention. This resulted in the formulation of Kyoto Protocol and the Paris Agreement. Another agreement was to “not to carry out any activities on the lands of indigenous peoples that would cause environmental degradation or that would be culturally inappropriate”.

The Convention on Biological Diversity was signed at the Earth Summit. It emphasized the protection of natural eco-regions and avoiding economically unsustainable growth. Despite its success, many of the agreements made in Rio such as fighting poverty and cleaning up our environment are yet to be accomplished.

Check Your Progress: 1

- Note:** 1) Use the space provided below for your Answers.
2) Compare your answers with those given at the end of the unit.

1. What is environmental law?

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2. What was the importance of the Rio conference?

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3. Explain Agenda 21.

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6.4 JOHANNESBURG TREATY

The World Summit on Sustainable Development, also referred to as the Earth Summit was held at Johannesburg, South Africa from 26 August to 4 September 2002. The United Nations convened the summit to discuss sustainable development. The summit brought together several leaders from business and non-governmental organizations. It was held 10 years after the first Earth Summit in Rio de Janeiro. The goal of WSSD was to hold a 10 year review of the 1992 UNCED to reinvigorate global commitment to sustainable development. It was therefore also informally nicknamed “Rio+10”. The main outcome of the Summit was the Johannesburg Declaration and the WSSD plan of implementation. A number of partnerships were launched to help implement commitments made at WSSD.

6.4.1 Plan of Implementation

It is conceived as a framework to implement the sustainable developments originally agreed at UNCED in 1992. The plan contains chapters on the following:

(i) Poverty eradication, (ii) consumption and production, (iii) the natural resource base, (iv) health, (v) small island developing states, (vi) Africa, (vii) other regional initiatives, (viii) means of implementation, and (ix) institutional framework. The Plan of Implementation contains over 30 targets, the majority of which have been established in the Millennium Declaration.

Some of the most significant targets include:

- i) To halve the number of the world’s poor living on less than \$ 1 a day 2015.
- ii) Significantly improve the lives of at least 100 million slum dwellers by 2020
- iii) To halve the proportion of people without safe drinking water by 2015.
- iv) To halve the proportion of people without access to sanitation by 2015.
- v) Cease destructive fishing practices and establish representative marine protective areas and networks by 2012,
- vi) Maintain or restore fish stocks to levels that can be sustainable harvested by not later than 2015,
- vii) Phase out chemicals with detrimental health impacts by 2020 and
- viii) Significantly reduce biodiversity loss by 2010.

6.4.2 Johannesburg Declaration on Sustainable Development

It did not set out international principles which could be invoked in legal or political contexts. It was a general philosophical contextualization for the more detailed commitments in the plan of implementation. It reaffirms political commitment to sustainable development and building a humane, equitable and caring global society. It highlights the integration of economic development, social development, and environmental protection, the three

pillars of sustainable development. It emphasises the need to eradicate poverty, change production and consumption patterns, and protect and manage the natural resource base. The declaration reaffirms the Millennium Development Goals, and welcomes decision taken at WSSD on targets, timetables and partnerships to improve access to clean water, sanitation, energy, healthcare, and food, and to protect biodiversity. It underscores the need for access by developing countries to financial resources, for opening of markets and the transfer of technology, and for the private sector to enforce accountability. It also addresses issues like armed conflict, terrorism, intolerance, combating communicable and chronic diseases. Finally, the Declaration reaffirms all countries commitment to the UN Charter and international law, calls for strengthening multilateralism and pledges to an inclusive process involving all groups.

Voluntary Partnerships: Partnership programs were launched in the form of voluntary agreements between one or more countries, international institutions, private sector and/or non-governmental organizations. More than 300 such partnerships were launched at the Summit, including 32 energy initiatives, 21 major water programs for biodiversity and ecosystem management.

6.5 SOME LEGISLATIONS OF INTERNATIONAL IMPORTANCE

Let us now learn about some more legislations of international importance. They include the following:

6.5.1 Cites

The increase in the trade in life forms and their products for commercial purposes beyond frontiers has been one major cause for the decline in the biodiversity. In order to prevent this, in the year 1973, CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) was convened. This multilateral treaty protects endangered plants and animals. This treaty was drafted as an outcome of the deliberations of the meeting of members of the International Union for Conservation of Nature (IUCN) in 1963. The convention was opened for signature in 1973. CITES came into existence on 1st July 1975. The primary objective was to curb the international trade in specimens of wild animals and plants in order to protect the survival of the species in the wild. Participation by member states to join this treaty is voluntary. Those nations that have agreed to be bound by the Convention are known as Parties. The parties to CITES are collectively referred to as the Conference of the Parties. They meet every 2 or 3 years to review the implementation. CITES is also attended by agencies of the UN, NGOs working in the relevant areas. CITES is legally binding on the Parties, but does not replace the existing national laws. It provides broad guidelines and the nations must design their own domestic laws to implement CITES. CITES Secretariat is administered by UNEP and is located at Geneva, Switzerland and all activities relating to the convention are co-ordinated.

Regulation of trade: CITES functions by regulating international trade in specimens of some identified species. Licensing system is followed for all import, export, re-exports covered by the Convention. Each Party to the Convention must designate one or more Management Authorities of administering the licensing system (Article IX of the Convention). Further, Scientific Authorities are to play advisory role on the effects of trade on the status of CITES-listed species.

Limitations: CITES focuses on trade at the species level. Unfortunately it does not deal with ecosystem approaches to conservation or destruction of habitat. CITES only protects charismatic species-mega fauna or animals - with high market value. CITES only controls their trade. It does nothing to hunting or killing of animals.

6.5.2 Vienna Convention

The ozone layer is that layer of the stratosphere, which protects the living beings on the earth from the harmful effects of ultraviolet rays from the sun. The widespread use and their release into the atmosphere of ozone-depleting compounds have increased levels of ultraviolet rays reaching the earth. In the Vienna Conference, 1985, the Vienna Convention for the Protection of the Ozone Layer was agreed upon. It is a multilateral environmental agreement and came into force in 1988. It is one of the most successful environmental treaties, ratified by 197 states as well as the European Union. The objectives of the Convention were to facilitate cooperation by research and information exchange on the effects of human activities on the ozone layer. It also aimed to adopt legislative or administrative measures against activities likely to have adverse effects on the ozone layer. It is the basis for international efforts to save the ozone layer. CFCs are the main chemical agents responsible for ozone depletion. This treaty does not include legally binding reduction goals regarding the use of CFCs.

6.5.3 Montreal Protocol

The Montreal Protocol on Substances that Deplete the Ozone Layer (a protocol to the Vienna Convention for the Protection of the Ozone Layer) is an international treaty for the protection of the ozone layer. Its objectives include phasing out the production of ozone depleting substances. It was agreed on 16 September 1987. It entered into force on 1 January 1989, followed by a first meeting in Helsinki, May 1989. Since then, it has undergone several revisions. It aims at the reduction in the production of CFC (substance) itself and not just emission of the CFCs. The parties are expected to reduce their consumption and production of CFC to the 1986 level, which is considered to be the base year. The protocol takes into consideration the different position of the developed and the developing countries. It has laid obligations on them, where the phasing out time is different. It requires phasing out all consumptions or production of most of the substances that deplete the Ozone layer by Jan 1996 by the developed countries and the consumption of HCFC to be frozen in 1996 and to be completely phased out by 2030. The complete elimination time that is fixed for the developing countries, unlike the developed countries is the year 2010

and the elimination period for Hydro Chlorofluorocarbons (HCFC'S) is 2040. Due to its universal acceptance, it is considered an environmental success story in the field of international co-operation.

6.5.4 Rotterdam Convention

The Rotterdam Convention (formally, the Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade) is a multilateral treaty to promote shared responsibilities in relation to importation of hazardous chemicals. The Convention was adopted in the year 1988. The Governments thought that here was a need to strengthen the procedure of acquiring prior consent for transboundary movement of certain hazardous substances to make it legally binding. It seeks to ensure that the hazardous chemicals are managed in an environmentally sound manner. The Convention also provides space for information exchange. The Convention empowers the national governments to decide on the entry of such hazardous chemicals into their country. The convention promotes open exchange of information. It calls on exporters of hazardous chemicals to use proper labelling, include directions on safe handling, and educate purchasers of any known restrictions or bans. Signatory nations can decide whether to allow or ban the importation of chemicals listed in the treaty, and exporting countries are obliged to make sure that producers within their jurisdiction comply.

6.5.5 Stockholm Convention on POP

Another important international environmental treaty is the Stockholm Convention on Persistent Organic Pollutants signed in 2001. The objective of this treaty is to eliminate or restrict the production and use of persistent organic pollutants (POPs) and became effective from May, 2004. This treaty is the first international legal instrument to focus attention on the dangers of persistent organic pollutants. These chemicals that are commonly used as pesticides in agriculture and to control insects causing diseases like malaria. The United Nations Environment Programme (UNEP) initiated for global action to be taken on POPs in 1995. POPs are chemicals that remain in the environment and have bio-accumulating properties in the food chains. They also cause toxicity to human health and the environment. The Intergovernmental Forum on Chemical Safety and the International Programme on Chemical Safety assessed the impact of twelve POPs popularly known as the dirty dozen. The parties agreed to phase out nine of the dirty dozen chemicals. They decided to use DDT for malaria control, and reduce inadvertent production of dioxins and furans. Parties to the convention have agreed to a procedure by which persistent toxic chemicals can be reviewed and added to the convention, provided they meet certain criteria for persistence and transboundary threat. The main feature of the Convention is that the developed countries provide new and additional financial resources and measures to eliminate production and use of intentionally produced POPs. They are also required to eliminate unintentionally produced POPs wherever possible, and handle POPs wastes in an eco-friendly way.

Precaution is exercised throughout the Stockholm Convention, with specific references in the preamble, the objective, and the provision on identifying new POPs.

6.5.6 Basel Convention

The international treaty regarding the control of transboundary movements of hazardous wastes and their disposal is known as the Basel Convention. The primary objective is to reduce the movements of hazardous waste between nations. Further, it specifically prevents the movement of hazardous wastes from developed to the lesser developed countries. This treaty does not include the movement of radioactive waste. This Convention also aims to reduce the amount and toxicity of waste generation, to ensure their safe disposal close to the source of generation. The Convention also aims to help in the safe disposal of the hazardous and other wastes generated by the lesser developed countries. The Convention was opened for signature on 22nd March 1989, and entered into force on 5th May 1992.

The key objectives of the Convention are as follows:

- i) Reduce transboundary movements of hazardous wastes and environmentally sound management of hazardous waste,
- ii) Provide for the treatment and disposal of hazardous wastes as close as possible to their sources of generation,
- iii) Minimize generation of hazardous wastes in terms of quantity and hazards.

The Basel Ban, adopted in 1994 by certain countries, bars shipments of hazardous waste from developed countries to less- developed countries.

6.5.7 Cartagena Protocol on Biosafety

The Cartagena Protocol on Biosafety to the Convention on Biological Diversity is an international agreement on biosafety. It is a supplement to the Convention on Biological Diversity effective since 2003. This Protocol aims to protect biodiversity from the potential threats resulting from genetically modified organisms. The Protocol outlines that precautionary principles should be applied for products that result from modern biotechnology. It permits developing countries to balance public health against economic benefits. In case there is insufficient scientific evidence regarding product safety it allows countries to ban imports of genetically modified organisms. It also requires exporters to label the shipments containing genetically altered commodities. This protocol is in accordance with the precautionary approach, contained in Principle 15 of the Rio Declaration on Environment and Development. The objective of the Protocol is to ensure an adequate level of protection for the safe transfer, handling and use of living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity. They also take into account human health risks, specifically focusing on transboundary movements.

The protocol defines a 'living modified organism' as any living organism that possesses a novel combination of genetic material obtained through the use

of modern biotechnology. Living organism means any biological entity capable of transferring or replicating genetic material. Modern biotechnology is defined in the Protocol to mean the application of *in vitro* nucleic acid techniques, or fusion of cells beyond the taxonomic family, that overcome natural physiological reproductive or recombination barriers. These are not the techniques used in traditional breeding and selection. 'Living modified organism (LMO) Products' are defined as processed material that are of living modified organism origin, containing detectable novel combinations of replicable genetic material obtained through the use of modern biotechnology. Some examples of LMOs are genetically modified agricultural crops for increased productivity and pest resistance. Examples of modified crops include tomatoes, cassava, corn, cotton and soybeans. Living modified organism intended for direct use as food or feed, or for processing are agricultural commodities from GM crops. Overall the term 'living modified organisms' is equivalent to genetically modified organism. The Protocol did not make any distinction between these terms and did not use the term 'genetically modified organism.' The Protocol applies to the transboundary movement, transit, handling and use of all living modified organisms that may adversely affect conservation and the sustainable use of biodiversity (Article 4 of the Protocol, SCBD 2000). The Protocol promotes biosafety by establishing rules and procedures for the safe transfer, handling, and use of LMOs, with specific focus on transboundary movements of LMOs. It features a set of procedures including one for LMOs that are to be intentionally introduced into the environment called the advance informed agreement procedure, and one for LMOs that are intended to be used directly as food or feed or for processing. Parties to the Protocol must ensure that LMOs are handled, packaged and transported under conditions of safety. Furthermore, the shipment of LMOs subject to transboundary movement must be accompanied by appropriate documentation specifying, among other things, identity of LMOs and contact point for further information. These procedures and requirements are designed to provide importing Parties with the necessary information needed for making informed decisions about whether or not to accept LMO imports and for handling them in a safe manner. The Party of import makes its decisions in accordance with scientifically sound risk assessments. The Protocol sets out principles and methodologies on how to conduct a risk assessment. In case of insufficient relevant scientific information and knowledge, the Party of import may use precaution in making their decisions on import. Parties may also take into account, consistent with their international obligations, socio-economic considerations in reaching decisions on import of LMOs. Parties must also adopt measures for managing any risks identified by the risk assessment, and they must take necessary steps in the event of accidental release of LMOs. To facilitate its implementation, the Protocol establishes a Biosafety Clearing-House for Parties to exchange information, and contains a number of important provisions, including capacity- building, a financial mechanism, compliance procedures, and requirements for public awareness and participation.

Check Your Progress: 2

Note: 1) Use the space provided below for your Answers.

2) Compare your answers with those given at the end of the unit.

1. What was the significance of the Johannesburg treaty?

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2. What did the Montreal protocol aim to achieve?

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3. Explain the significance of Cartagena protocol on biosafety.

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6.6 LET US SUM UP

In this unit we have studied about the need for environmental law and its evolution. The very significant environmental conferences and their achievements were discussed. Some important international environmental law agreements like CITES, Cartagena protocol on biosafety, Stockholm convention on POP's, Montreal protocol, Vienna convention, Rotterdam convention were discussed.

6.7 KEY WORDS

POP's POP's: They are persistent organic pollutants (POPs). These chemical substances that persist in the environment, bio-accumulate through the food web, and cause adverse effects to human health and the environment.

The Biosafety Protocol: It aims to protect biological diversity from the potential risks posed by genetically modified organisms, resulting from modern biotechnology.

Hazard Hazard: It can be defined as a potential threat to humans and their welfare and risk as the probability of hazard occurrence.

Living modified organism (LMO) Products: They are defined as processed material that are of living modified organism origin, containing detectable novel combinations of replicable genetic material obtained through the use of modern biotechnology.

IPCC: Intergovernmental Panel on Climate Change.

6.8 FURTHER READINGS

Tiwari, H.N. 2008. Environmental law. Allahabad Law Agency. 548 p.

Jaswal, P.S and Nishtha Jaswal. 2011. Environmental Law. Allahabad Law Agency. 612 p.

6.9 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Your answer should include the following points:

- Global environmental issues are examined under the subject of International Environmental law. Debates over environmental issues are based on the principles of international law, numerous international agreements and declarations. International Environmental law is not a separate branch of law but a part of the International law which governs relations between states.
- Environmental international law has evolved from a number of sources. The most important source is the customary international law. Customary international laws are the norms followed by most countries as a matter of custom. They are responsible for the common thread that binds all countries. Other sources of the international law include general principles of law recognized by civilized nations.
- International law is also evolved from subsidiary sources such as decisions of courts and tribunals and writing of jurists.

2. Your answer should include the following points:

- The Rio conference was held 20 years after the first International conference on environmental issues at Stockholm. The main objective of the Rio conference was to stop environmental degradation of the planet. Following the Stockholm conference, UN in the year 1983 set up a Commission on Environmental and Development led by Brundtland from Norway. The Commission suggested the concept of Sustainable Development. This is defined as “the development which meets the needs of the present generation without compromising the ability of the future generations to meet their own needs”. Based on the Brundtland report, 1987- the UN General Assembly called for the convention of UNCED (United Nations Conference on Environment and Development). UNCED was held at Rio De Janeiro, Brazil in June, 1992 represented by 172 governments. Several representatives of non-governmental organizations (NGOs) attended. It focussed on balancing development that support socio-economic development and avoids damage to the environment. It also stressed on the need to bring about global partnership to tackle the issues.
- Issues covered

- Agenda 21
- Important agreements

3. Your answer should include the following points:

- Agenda 21 is a dynamic programme and a voluntary action plan. It is called so because it is a comprehensive blue-print for local, national, regional and global actions. The proposed actions will enable the transition to sustainable development in the 21st century. It was adopted by more than 178 Governments at the Rio Conference. It comprises of 40 chapters and hundreds of programme areas. Agenda 21 lays down the ways and means in the reduction of wasteful and inefficient consumption patterns. It recognized the need of developed countries like extending financial support and aiding technology transfer to developing countries to achieve sustainable development.

Check Your Progress: 2

1. Your answer should include the following points:

- The World Summit on Sustainable Development, also referred to as the Earth Summit was held at Johannesburg, South Africa from 26 August to 4 September 2002. The United Nations convened the summit to discuss sustainable development. The summit brought together several leaders from business and non-governmental organizations. It was held 10 years after the first Earth Summit in Rio de Janeiro. The goal of WSSD was to hold a 10 year review of the 1992 UNCED to reinvigorate global commitment to sustainable development. It was therefore also informally nicknamed “Rio+10”. The main outcome of the Summit was the Johannesburg Declaration and the WSSD plan of implementation. A number of partnerships were launched to help implement commitments made at WSSD.

2. Your answer should include the following points:

- The Montreal Protocol on Substances that Deplete the Ozone Layer (a protocol to the Vienna Convention for the Protection of the Ozone Layer) is an international treaty for the protection of the ozone layer.
- Its objectives include phasing out the production of ozone depleting substances. It was agreed on 16 September 1987. It entered into force on 1 January 1989, followed by a first meeting in Helsinki, May 1989. Since then, it has undergone several revisions. It aims at the reduction in the production of CFC (substance) itself and not just emission of the CFCs. The parties are expected to reduce their consumption and production of CFC to the 1986 level, which is considered to be the base year.
- The protocol takes into consideration the different position of the developed and the developing countries. It has laid obligations on them, where the phasing out time is different. It requires phasing out all consumptions or production of most of the substances that

deplete the Ozone layer by Jan 1996 by the developed countries and the consumption of HCFC to be frozen in 1996 and to be completely phased out by 2030. The complete elimination time that is fixed for the developing countries, unlike the developed countries is the year 2010 and the elimination period for Hydro Chlorofluorocarbons (HCFC'S) is 2040.

3. Your answer should include the following points:

- The Cartagena Protocol on Biosafety to the Convention on Biological Diversity is an international agreement on biosafety. It is a supplement to the Convention on Biological Diversity effective since 2003. This Protocol aims to protect biodiversity from the potential threats resulting from genetically modified organisms. The Protocol outlines that precautionary principles should be applied for products that result from modern biotechnology. It permits developing countries to balance public health against economic benefits. In case there is insufficient scientific evidence regarding product safety it allows countries to ban imports of genetically modified organisms. It also requires exporters to label the shipments containing genetically altered commodities. This protocol is in accordance with the precautionary approach, contained in Principle 15 of the Rio Declaration on Environment and Development. The objective of the Protocol is to ensure an adequate level of protection for the safe transfer, handling and use of living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity. They also take into account human health risks, specifically focusing on transboundary movements.

UNIT 7 ENVIRONMENTAL SUSTAINABILITY

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- 7.7 Check Your Progress: Possible Answers

7.0 INTRODUCTION

The idea of sustainable development has been there, ever since human beings started using natural resources for their development. All forms of development, be it economic, educational, technological, agricultural, or social, have taken a toll on the consumption of environmental resources. Sustainable development talks about a development scenario where resources are consumed so that humans' needs are not compromised in the present, and resource availability is also ensured for the future. Since the beginning, we have consumed natural resources, but the consumption rate has increased exponentially after the industrial revolution.

On an international platform, the idea of environmental equity and sustainable use of resources was first discussed in the **United Nations**

Conference on the Human Environment (UNCHE), held in Stockholm, Sweden, in 1972. For the first time, the representatives of 114 countries attended a conference that brought the environment to the forefront and established that economic development is not independent of environment conservation (Sharma, P.D., 2016). The declaration of the United Nations Conference on Human Environment recognized the finite nature of resources available on earth and emphasized its judicious use to remain available for future generations. It also recognized that developmental needs are different for developing and developed nations. A large population in developing countries is living below a decent standard of living required in food, clothing, shelter, and health. The developing countries must strive to meet the needs of the population and, at the same time, manage the resources well. On the other hand, the developed nations need to balance the resource consumption pattern and increasing demands (https://www.ipcc.ch/apps/nj-lite/srex/nj-lite_download.php?id=6471).

In 1983, the UN General assembly established a commission, which later came to be known as the World Commission on Environment and Development (WCED), to come up with a report on the state of the environment and propose sustainable development to the year 2000 and beyond. The Commission, also known as the Brundtland Commission, named after its Chairman Gro Harlem Brundtland, the former Prime Minister of Norway, came up with a report in 1987. The report titled “Our Common Future” discussed various environmental challenges like food security, biodiversity conservation, industrial pollution, energy crisis, urbanization, and possible solutions. It also defined sustainable development as *“the development that meets the needs of the present without compromising the needs of the future generations to meet their own needs”* <https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf>.

Sustainable development is a broad term that incorporates the concept of environmental systems’ carrying capacity and that economic development is possible only through resource conservation. For an idea to be sustainable in the long run, it has to be successful on three parameters, i.e., **Environmental viability, Economic feasibility, and Social acceptability** (Wright & Boorse, 2015). These are also known as the three pillars of sustainability (Fig. 7.1)

- **Environmental Viability:** This aspect focuses on developing ideas and products that put the least risk on the environment and minimizes resource consumption. The life-cycle assessment of any product can describe its production, usage, and impact as a waste dumped on the environment. The products with lower water and carbon footprints can be sustained environmentally for a longer period.
- **Economic Feasibility:** From the business and profitability perspective, economic viability/profitability is the most important driving force. In the long run, a business pattern in coherence with the concept of environment conservation is likely to last longer successfully. The industrial sector anywhere in the world relies on natural resources, e.g., coal, freshwater, wood, metals, minerals, etc. They all are essential raw materials for major industrial processes. Ensuring their availability, in

the long run, is crucial to the health of the industry. Whenever the supply starts reducing, the manufacturing and processing cost increases, reflected in its final pricing.

- **Social Acceptability:** For any environmentally sustainable idea to be successful, it must be accepted and adopted. The social acceptance comes with the convenience of use, appropriate pricing, upliftment of the local, indigenous talent, and resources. All the stakeholders in the society, employers, employees, consumers have to be satisfactorily involved.

7.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- discuss the concept of sustainable development;
- discuss the steps to achieve sustainable development through specific targets and indicators;
- discuss the international efforts towards sustainable future for climate change, biodiversity conservation and movement of hazardous substances; and
- identify different ethical viewpoints towards environment.

7.2 ENVIRONMENTAL SUSTAINABILITY: TARGETS AND INDICATORS

As discussed earlier, sustainable development is achieved when all three pillars stand strong (Fig. 7.1). The environmental viability pillar, in particular, focusses on ecosystems and the support they provide to humankind. Achieving a sustainable solution must involve an understanding of economic processes (procuring raw material, production, and disposal of waste) and its impact on the environment and health (Miller Jr, G.T., 2007).

Sustainable Development

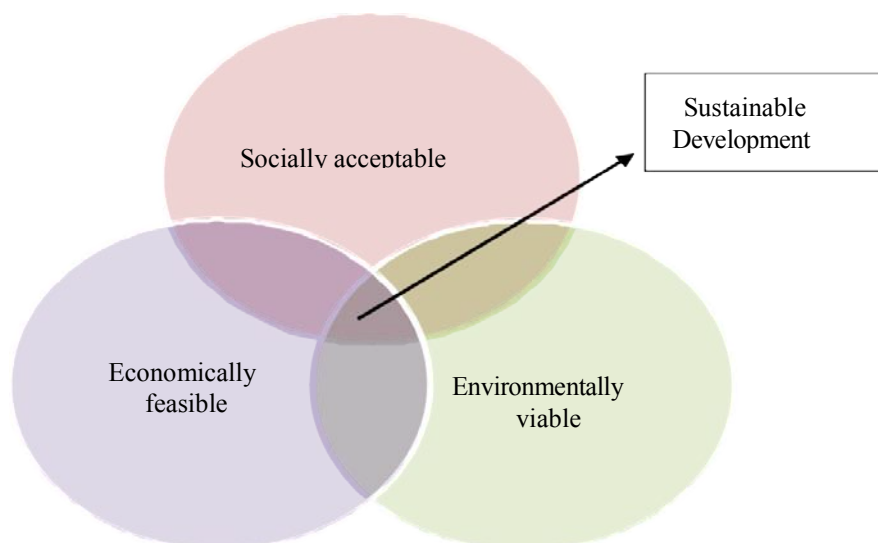


Figure. 7.1: Sustainable solutions: where the common concerns of environment, economics, and society overlap. Source: Wright and Boorse, 2015.

To objectively identify and describe environmental sustainability, certain goals and targets are set, and the changes these targets would reflect on the environment are measured using specific indicators. All the targets underline the fundamentals of environmental conservation and minimal harm to society. From time to time, the world has come together to incorporate environmental values in the economic system. Some of the targets that have played a defining role in the world's modern environmental history are discussed here.

7.2.1 Millennium Development Goals

In the year 2000, the United Nations Millennium summit adopted certain goals known as Millennial Development Goals (MDGs) to be achieved by 2015 (Fig 7.2). The eight MDGs and seventeen targets under them have specific indicators to measure their success (<https://www.un.org/millenniumgoals/>). Table 7.1 gives MDGs and the targets assigned under environmental sustainability.



Fig 7.2: Millennium Development Goals (MDGs).

Table 7.1: The Millennial Development Goals, targets, and indicators (Figures in 1990 are compared to that in 2015).

Goal 1	Eradicate extreme poverty and hunger
Goal 2	Achieve universal primary education
Goal 3	Promote gender equality and empower women
Goal 4	Reduce child mortality
Goal 5	Improve maternal health.
Goal 6	Combat HIV/AIDS, malaria, and other diseases
Goal 7	Ensure environmental sustainability.
Targets for Goal 7	Integrate the principle of sustainable development into country's policies and programs, and reverse environmental resource loss. Indicator: The proportion of land area covered with forests. Consumption of ozone-depleting substances. The proportion of terrestrial and marine areas protected.

	Reduce by half the proportion of people without sustainable access to safe drinking water and basic sanitation. Indicator The proportion of people using improved drinking water sources.
Goal 8	The proportion of people using improved sanitation facility. Achieve a significant improvement in the lives of at least 100 million slum dwellers by 2020. Indicator: The proportion of the urban population living in slums. Forge a global partnership for development.

Source: <http://mdgs.un.org/unsd/mdg/Host.aspx?Content=Indicators/OfficialList.htm>.

The Millennium Development Goals targets were partially met in terms of children's education (goal 2), decreasing mortality rate (goal 4), and removing extreme poverty (goal 1). However, the target that was far from being successfully implemented was goal 7 on environmental sustainability (Fehling et al., 2013). The goals' incomplete success can be linked to limitations like inadequate resources, lack of focus and accountability, and ambiguity on how the targets were designed (Fehling et al., 2013). Carrying on the momentum developed during MDGs implementation, a set of 17 new goals known as Sustainable Development Goals (SDGs) were designed to be fulfilled in the next 15 years till 2030.

7.2.2 Sustainable Development Goals

To achieve sustainable development, addressing all the environmental challenges the world is facing today, a comprehensive plan to implement sustainability is a must. To achieve that, United Nations member states adopted seventeen goals known as Sustainable Development Goals (SDGs) (Fig 7.3) in 2015. The goals are quite comprehensive, covering all the aspects of development. These seventeen goals strive for sustainability, taking up different social and environmental causes, and successfully attaining them by 2030. They are as follows:



Fig 7.3: The Sustainable Development Goals adopted in 2015 with a target to be achieved by 2030.

1. **No poverty:** Much progress has been made on eradicating poverty since 1990, but the poverty figures indicate that a large population of the world is still poor. Population living on wages of less than 2\$US/day is huge in South Asia. It is important to address poverty imperatively as it is mostly the poor who are more vulnerable to climate change impacts.
2. **Zero Hunger:** With the development of agricultural technology, food production has increased, and the number of **people** dying of hunger has decreased drastically. Food security remains a major challenge, as the quality parameters of food are not met. In many developing countries, malnutrition is affecting millions of children and adults.
3. **Good health and wellbeing:** This goal is to achieve good health for all and wellbeing at all ages. Good health is primary for the development of a healthy society. A large population in developing countries does not have access to health services and insurances. Much progress has been made in increasing life expectancy and medicines availability for chronic and life-threatening diseases like diabetes and blood pressure, etc., yet, a lot is left to achieve. The goal would be fulfilled only when the health services reach the poorest of the poor.
4. **Quality education:** Aims to achieve quality and inclusive education from learning and obtain skills for a lifetime providing job opportunities.
5. **Gender equality:** Much work that women do in the formal and informal sectors goes unnoticed, as women are either not paid or paid less than their male counterparts. This goal is set to achieve gender equality for women in all spheres of life. The work done by women can contribute substantially to the economic growth of the country.
6. **Clean water and sanitation:** There is a large part of the world's population that do not have access to clean drinking water, and many more do not have access to clean sanitation services. Both are primary requirements for a healthy life. Lacking these, people are subjected to many water-borne diseases. Huge infrastructural investments are needed to provide clean drinking water and sanitation services to all.
7. **Affordable and clean energy:** Energy is an essential requirement for development in the industrial, agricultural, and domestic sectors. In India, we have been dependent on energy from non-renewable sources for a long time. The need of the hour is the availability of energy for all generated in an environment-friendly manner. Non-conventional sources hold the key to the future of energy. India has taken a key step towards developing solar energy through the Ministry of New and Renewable Energy, giving subsidies to set up solar power plants. International Solar Alliance, to increase the use of solar energy through solar panels is a great commitment of India towards a sustainable future.
8. **Decent work and economic growth:** This goal aims to promote sustainable economic growth and simultaneously provide decent working conditions. This also tries to support technological innovations that are environmentally friendly and save resources.

9. **Industry innovation and infrastructure:** This goal promotes technological innovations to address problems like urbanization, mass transportation, cheaper renewable energy, etc., and even internet connectivity that is crucial in the spread of information and knowledge.
10. **Reduced inequalities:** This goal intends to reduce the economic inequality that exists in society. A top few percent of the population owns or governs the majority of the resources. This inequality is less in developed countries and high in developing countries. The ecological footprint data from across the world highlights this disparity.
11. **Sustainable cities and communities:** This goal is about making our cities and urban areas more sustainable. With technological development and industrialization on the rise, urban sprawls come up and spread very fast. A large population of the world would be living in urban areas than rural by 2050. To make those cities able to thrive populations, growth pattern has to be sustainable.
12. **Responsible consumption and production:** Natural resources on earth are limited, limiting the economic growth rate. To keep a balance between resource availability and meeting the entire population's basic needs, sustainable consumption, and production hold the key.
13. **Climate action:** Climate change is an environmental emergency experienced by all the world countries to a different extent. The effect comes from an increase in the earth's average temperature and frequent recurrence of natural calamities. Mostly, it is the poor that bear the brunt of it. Huge infrastructural investment in shifting to lower carbon emissions, industrial processes and energy production is must to keep climate change under control.
14. **Life below water:** Oceans are major carbon reservoir and host to a great biodiversity. Fish and other seafood are a major source of food throughout the world. Fish stocks are too over-exploited to maintain a sustainable population. Mechanized fishing, pollution, climate change, ocean acidification is harming biodiversity like never before. This goal aims to manage marine and coastal systems sustainably.
15. **Life on land:** land and forest resources are extensively utilized and diverted for urbanization, industrialization, agricultural uses, and roads. Biodiversity conservation is a must to maintain the ecosystem processes, food chain, and prevent further degradation of land leading to desertification.
16. **Peace justice and strong institutions:** The three pillars of sustainable development would only be fulfilled when there is peace and functioning law and order situation in the community. Disturbance, gender inequality, insecurity, and violence in the community hampers the region's economic growth and prosperity. Strengthening of institutions for the establishment of peace and security in society is essential for complete sustainability.

17. **Partnerships for goals:** Environment sustainability is a comprehensive process that requires coordination between developed and developing countries to cope better with natural disasters and technological exchange for environment-friendly industrial production and processes. According to the United Nations Conference on Trade and Development investments, trillions of dollars are required to accomplice SDGs.

All these goals may seem independent but work holistically to make an environmentally sustainable state. A comprehensive set of indicators (global indicator framework) is developed and updated regularly to measure these goals' implementation status. The targets' indicators are divided into subcategories based on age, sex, ethnicity, migratory status, or other categories. The global indicator framework has 247 indicators, out of which 231 are unique, while a few are repeated under multiple goals as they encourage and indicate comprehensive growth of the society. The environment-based goals like access to energy (goal 7) measure the population's proportion with primary reliance on clean energy sources (indicator 7.1.2). Goal 12 on sustainable production and consumption considers material footprint, material footprint per capita, and material footprint per GDP (indicator 12.2.1). For the goal to take urgent action to combat climate change, indicators like the number of lives lost in natural disasters (13.1.1) and the measures taken for disaster risk reduction on the lines of Sendai framework, 2015-2030 (13.1.2) are there. Goal 15 of promoting sustainable ecosystems has indicators like the proportion of forest cover for the total land area (15.1.1), coverage of protected sites to conserve mountain biodiversity (15.5.1), and the proportion of traded wildlife that was poached or illicitly trafficked (15.c.1). The link for the latest set of indicators, upgraded in 2020, is given here ([https:// unstats.un.org/sdgs/indicators/indicators-list/](https://unstats.un.org/sdgs/indicators/indicators-list/)).

7.2.3 Life Cycle Assessment

Apart from the above measurable inputs for SDGs, there are indicators like Life Cycle Assessment (LCA) that “quantifies all relevant emissions and resources consumed, the related environmental and health impacts and the depletion of the resource associated with any product or services”(EC-JRC, ILCD Handbook, 2010). A thorough life cycle analysis of products and services needs to be carried out based on the definition. With detailed information about resource consumption and the product/service's impact on the environment, the consumer/local bodies and governments can make informed decisions that may increase sustainability. LCA aims to compare alternatives, impacting the environment in relative terms (Dong and Hauschild, 2017). The impacts on health and the environment can be local/regional and may vary over the temporal scale. The financial concerns, along with the environment, can be a guiding principle for further usage.

Check Your Progress: 1

Notes: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Why is there a need of sustainable development?

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2. What are the three pillars of sustainable development?

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7.3 NATIONAL AND INTERNATIONAL CONVENTIONS

Efforts on the national and international fronts are required for sustainability to survive in the long run. Internationally, many conventions are held, treaties and protocols are signed to better society and conserve the environment. Some of the major conventions and protocols are discussed here.

7.3.1 Earth Summit 1992

The United Nations Conference on Environment and Development (UNCED), also known as Earth Summit/Rio Summit held in Rio-de-Janeiro, Brazil, from June 3 to June 14, 1992, is considered one landmark conference. The idea of environmental conservation, sustainable development, and the efforts to bridge the gaps between environment and economy were discussed internationally. After the Stockholm conference in 1972, the Earth Summit was a step forward in taking concrete actions on the ground. Leaders from 170 countries worldwide, including 115 heads of the states, participated in the event. The conference was held to bring the member states on a common platform to achieve a bigger sustainability objective. The conference's major outcomes include the Rio Declaration, Convention on forestry, also known as forest Principles, and Agenda 21. Some legally binding agreements like the United Nations Framework Convention on Climate Change (UNFCCC), Convention on Biodiversity, and Convention to Combat Desertification were also opened for signature to the member states. These are discussed briefly below:

Rio Declaration (Earth Charter): The declaration states 27 guiding principles for the conservation and restoration to maintain different ecosystems. The document was signed by all the heads of the States and Government representatives. It mentions the "Polluter Pays Principle," which makes the polluter responsible for the damage caused to the natural environment and bear the clean-up cost. When we start to form a common baseline, it is observed that developed countries have caused a lot more pollution than developing countries. The cost they have to bear for clean-up

is huge. The developing countries, which are still fulfilling basic requirements for their countrymen, may pollute in the time to come (Sharma, 2015). The great outcome for bridging these gaps was the principle of “Common but differentiated responsibilities” to conserve ecosystems.

Convention on Forestry or Forestry Principles: It is a document that lays down forest conservation and sustainability principles. Initially, there was a lot of controversy over this issue as developed nations, like the U.S.A., Germany insisted on having an internationally binding agreement for global conservation and forest management. This would help them globalize the forest resources of developing nations and use them for industrial processes. Countries like India were principally against this as forests are a resource and part of tradition (Sharma, 2015). It remains a non-legally binding agreement.

Agenda 21: Agenda 21 is a comprehensive document that lays down the principles of sustainable development covering all the aspects of society and came with a concrete plan for the implementation. It addresses the contemporary environmental problems and the social aspects aggravating poverty, social injustice, consumption patterns, human health conditions, etc. The agenda explicitly discussed the financial mechanisms and partnerships among the countries to achieve the goals mentioned above. It postulated that partnerships should be such that developing countries should get appropriate financial and technological assistance to meet sustainable growth targets. Environment- friendly technology would enable the developing countries in the long run by reducing the damage done to the environment. Countries receiving any form of international assistance has to work out a detailed analysis of the existing situation and the basis on which the funding is to be given, like determining the research areas where and how the changes in the environment affect the people, understanding social aspects, ways to alleviate poverty, providing health care facilities for all, etc. and extensive regular monitoring of situation to account for improvement over time.

Overall, the Earth Summit was a successful convention in formulating the idea of earth conservation and taking it one step ahead on a global platform except for a few setbacks where developed countries did not successfully contribute financially for technological assistance to developing countries. Not enough funds were collected through the Global Environment Facility (GEF). No specified limits of pollution reduction were set to achieve within a time frame of 8-10 years. It requires a great deal of commitment on the part of the countries to meet the designated targets without compromising development.

7.3.2 United Nations Framework Convention on Climate Change (UNFCCC) and the Kyoto Protocol

The United Nations Framework Convention on Climate Change is the first UN body involved in making policy to prevent climate change. It is a parent treaty of the Kyoto Protocol of 1997 and the Paris agreement of 2015 (<https://unfccc.int/about-us/about-the-secretariat>). Under UNFCCC, annual meetings of member countries are held. It helps its 197 member states decide upon plans and policies to mitigate existing climate change and prevent

further increase. It also helps them in devising new mechanisms for achieving the same goal.

Kyoto Protocol: The Kyoto Protocol was adopted in Kyoto, Japan, in December 1997 and came into force in February 2005. The protocol was set to reduce the emission of six major Green House Gases (GHG) like carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluor-o-carbons (PFCs), and sulfurhexafluoride (SF₆). The protocol also encouraged financial mechanisms to achieve the targets through a carbon-credit system and coming up with non-renewable energy sources with fewer carbon emissions. It guided developed countries to provide

funds for new technological implementation in developing countries yet to install and develop it. The percent reduction targets were different for developed countries under the principle of common but differentiated responsibilities.

Paris Agreement: In 2015, a major follow up agreement was signed by the member countries having to relook on the existing targets and reformulate them. It became more specific by setting the average global temperature limit to well below 2.0oC above pre-industrial levels. In this agreement, the member countries set their own Nationally Determined Contributions (NDCs) to prevent climate change. It came into force on 4th November 2016. Only 189 countries out of 197 member nations have ratified the agreement. India has committed herself to the cause of the prevention of climate change and working hard to achieve the targets. The United States of America, one of the world's largest economies, walked out of this agreement during the Paris meeting. India remains committed to its cause of the environment.

7.3.3 Convention on Biological Diversity

The Convention on Biological Diversity (CBD) is an international agreement to conserve biodiversity, use its products sustainably, and share the benefits of utilizing biological, genetic resources. The convention came into force on December 29, 1993, and was ratified by 196 member nations (<https://www.cbd.int/>). The Conference of Parties of the Convention, in due course of time, have developed important supplementary treaties contemporary to developing science and technology. Two such treaties are Cartagena Protocol and Nagoya Protocol.

Cartagena Protocol: The Cartagena Protocol on Biosafety to the Convention on Biodiversity. It was adopted on January 29, 2000, in Montreal, Canada (originally scheduled for Cartagena, Quebec) and came into force on 11th September 2003. A total of 173 of the member nations have ratified it. The protocol is about the movement of Living Modified Organisms (LMOs) across the countries as it can be a potential risk to any country's native biodiversity. The biosafety protocol encourages countries to declare complete information about such organisms to make an informed decision by the importing countries.

Nagoya Protocol: The Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to

the Convention on Biodiversity. The protocol provides a framework for fulfilling one of the objectives of CBD. It was adopted on 29th October 2010 in Nagoya, Japan, and came into force on 12th October 2014. Providing legal clarity and transparency and distributing benefits of utilizing genetic resources fairly is the protocol's main aim. It also details the mechanisms at national levels to attain the objectives.

Check Your Progress: 2

Notes: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Why the Earth Summit, 1992 is considered a landmark conference in the field of environment?

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2. Which international protocol work towards the prevention of climate change?

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7.3.4 United Nations Convention to Combat Desertification (UNCCD)

The UNCCD is an international legally binding agreement focussing on preventing land degradation and achieving sustainable land management ([https:// www.unccd.int/](https://www.unccd.int/)). The Convention was adopted on 17th June 1994 in Paris and came into force in December 1996. In all, 197 member parties have ratified it. The convention is strategic towards preventing land degradation through natural processes as the drylands (arid, semi-arid, and sub-humid areas) lose soil productivity and tend to expand, which may lead to desertification over time. In its 2018-2030 target, the convention is committed to achieving Land degradation Neutrality (LDN) to restore degraded lands' productivity.

7.3.5 Vienna Convention on the Protection of the Ozone Layer

The ozone hole over the South Pole was first discovered in 1985 by the British Antarctic Survey team led by Joseph Farman. The absence or reduction of ozone in the stratosphere can have harmful impacts on humans, flora, fauna, and other essential ecosystems on earth. Soon after discovering the ozone hole, an international convention was held in Vienna, Austria, in 1985, laying a framework for protecting the ozone layer by reducing the emission of Ozone Depleting Substances like chlorofluorocarbons (CFCs). The convention was adopted on 22nd March 1985 and came into force on 22nd September 1988.

Montreal Protocol: The Montreal Protocol under Vienna convention specifies phasing out production of Ozone Depleting Substances (ODS) to protect the ozone layer present in the stratosphere. It came into force on 16th September 1987, and in the year 2012, it became the first treaty to be universally ratified by all the countries of the world (<https://www.environment.gov.au/protection/ozone/montreal-protocol>). The protocol has a comprehensive list of ODS like chlorofluorocarbons(CFCs), halons, hydrochlorofluorocarbons(HCFCs), and hydrofluorocarbons (HFCs), etc. to be phased out with deadlines. The deadline may vary for developed and developing countries based on the principle of common but differentiated responsibilities.

7.3.6 Convention on Migratory Species (CMS)

The convention, also known as the Bonn Convention, was adopted in 1979 and came into force in 1983. There are 117 parties to the agreement (<https://www.cms.int/>). The convention on the conservation of migratory species lays down a framework for the conservation of wildlife species (terrestrial, aquatic and avian) that migrate from one region to another to complete their life cycle. The states through which the migratory species pass are called the Range states, and a strong commitment on the part of the states is required to protect them as long as their stay in foreign territories. In order to achieve this, it coordinates not only with the states but NGOs and other partners as well. A list of species threatened for extinction is provided in its Appendix, and the convention encourages its range states to coordinate for the successful protection of the species.

7.3.7 International Union for Conservation of Nature (IUCN)

It is a membership union of around 1400 member organizations across the world. The member's organizations include governments, wildlife organizations, NGOs, and other stakeholders working on environmental conservation (<https://www.iucn.org/about>). The variety of members worldwide help IUCN pool data and information required to conserve the environment and use natural resources sustainably. The organization was found in Fontainebleau, France, in 1948. IUCN collaborates with different organizations and guides governments at the national level to shape the biodiversity conservation strategy. The IUCN's in-situ protection status categories help in establishing national parks, wildlife sanctuaries, and biosphere reserves. The IUCN's **red list** is a comprehensive list of flora and fauna facing different levels of extinction threat. It also works closely with UNESCO in the conservation of natural heritage sites. IUCN helps conventions like CBD, CITES and Ramsar Convention, etc., in implementing biodiversity conservation.

7.3.8 Convention on International Trade of Endangered Species of Wild Flora and Fauna (CITES)

It is an international agreement between member nations to check on international trade of wildlife species of plants and animals so that their existence is not threatened. <https://www.cites.org/eng/disc/what.php>. The

CITES resolution was adopted in 1963 and came into force on 1st July 1975. The convention maintains a check on live animals and plants' trade and a wide range of products derived from them, e.g., exotic leather and furs, timber products, elephant tusks, tiger skin, other body parts, etc. CITES ensure that the trade of any such plants, animals, and products is authorized through a licensing system, and scientific authorities are guiding on the extent of trade. Convention has an extensive list of species covered under appendix I, II, and III and the levels to protect them from exploitation. To date, over 35000 species of plants and animals are listed under CITES. Not all the species listed are endangered or threatened in existence but enlisting them makes the sustainable use of natural resources more viable for the future.

7.3.9 Ramsar Convention on Wetlands

The Ramsar Convention on Wetlands (waterfowl convention) is a framework convention for conserving wetlands and wise use of its resources. The Convention was adopted on 2nd February 1971 in the city of Ramsar, Iran, and came into force in 1975 (<https://www.ramsar.org/about/>). The convention uses the term wetlands broadly and includes water bodies like lakes, rivers, deltas, mangroves, coastal areas, and man-made bodies like rice fields, reservoirs, etc. In order to become a member of the convention, the country has to designate at least one site of ecological importance in the country and cooperate to manage the transboundary wetlands. To date, there are 171 member parties to the convention. The convention works in close association with six other International Organisation Partners (IOPs) working on nature conservation that includes IUCN, World Wide Fund for Nature (WWF) International, Wetlands International, Wildfowl and Wetlands Trust and International Water Management Institute (IWMI) and Birdlife International for proper management of wetlands. India has 37 Ramsar sites (wetlands of international importance) that include some of the major lakes like Chilika Lake in Odisha, Wular Lake in Jammu and Kashmir, Hussainsagar Lake in Hyderabad, and others. Under the convention, a register is maintained where a change in any Ramsar site's ecological character has occurred because of natural or human activities, and the record is referred to as Montreux Record.

7.3.10 Basel Convention on Transboundary Movement of Hazardous Substances

It is a convention to check on the movement of hazardous substances across borders and prevent it from being disposed of in any particular region. The convention aims to reduce the generation of hazardous waste and ensure its disposal in an environmentally sound manner. In its annexures I, VIII, and IX, it classifies and describes two types of waste: hazardous waste and 'other wastes that require special consideration for management and disposal. The convention was adopted on 22nd March 1989 in Basel, Switzerland, and came into force on 5th May 1992 (<http://www.basel.int/theconvention/overview/tabid/1271/default.aspx>). The transfer of hazardous waste can only occur between countries as a bilateral agreement, even when one of the countries may not be a party to the convention.

The convention also provides technical assistance for the management and disposal of wastes, especially to developing countries, to prevent its impacts on people's health.

7.4 ENVIRONMENTAL ETHICS

Ethics is a philosophical term based on the morals of what humans believe is right and wrong. Environmental ethics describes the views of human beings towards the environment. (<https://plato.stanford.edu/entries/ethics-environmental/>). Broadly, how we perceive the environment, ecosystems, and natural resources can be classified into three categories.

7.4.1 Anthropocentric View

It is a belief that puts humans, their needs at the focus and considers human beings the most important and dominant species on earth. It believes that all the earth's resources belong to humans, and we can manage them. Some authors (Miller, Jr., G.T., 2007; Miller, and Spoolman, 2013) explain it as a human-centered and life-centered world view. It further explains that we would never run out of resources because of technological development and management, leading to the possibility of unfettered economic growth. This view considers that all the natural resources on earth belong to humans.

7.4.2 Bio-centric View

It states that human beings and other creatures of the world are part of nature and all are equal life forms. All the creatures of earth exist in interdependence with each other to maintain balance. This theory also suggests that we, as humans, have a moral responsibility towards fellow life forms. Conservationist Aldo Leopold, in an essay "Land Ethic" from the book 'A Sand County Almanac' published in 1949, described human beings' responsibility to protect wilderness as part of the ethical treatment of fellow life forms (Misra and Pandey, 2018; Wright and Boorse, 2015). This view is sometimes also referred to as Environmental Stewardship, which highlights that we should not waste natural resources and our success depends on how well we manage earth's life support system for us and the rest of nature (Miller, 2007).

7.4.3 Eco-centric view

It states that the environment and the nature in which we exist are far wider and more creative on their own. Humans are only one of the species found on earth and exist as part of the larger community. It acknowledges and appreciates the fact that every species has an intrinsic value attached to it. Every species plays an important role in the functioning of natural processes and is valuable for the ecosystem.

Environmental sustainability is intricately linked to environmental ethics as far as the conservation of resources for future generations is concerned (Sulphey and Safeer, 2017). The environmental ethics we follow today, says a lot about our sustenance on earth, and certain ethics have been part of our tradition for a long time. The existence of Sacred Groves in different parts of

our country shows that conservation to protect all the essential ecosystem functions is not a new concept for us. Religions like Hinduism, Islam, Jainism, Buddhism, and Sikhism, all preach to conserve the environment (Misra and Pandey, 2018). Development accompanied by the code of ethics can take us a long way on the road to environmental sustainability.

Check Your Progress: 2

Notes: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. How environmental ethics and culture help in environmental protection?

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2. What are the different viewpoints towards environment?

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7.5 LETS US SUM UP

In this unit, we discussed the concept of sustainable development and the essential criteria to make it applicable. Various goals, like, the Millenium Development Goals and the Sustainable Development Goals are set from time to time along with their indicators to check its implementation status. As the idea of sustainable development is holistic and comprises of environmental conservation and social justice, it can be successful only when the whole world takes action in a concerted manner. With the help of international organisations like United Nations, many international conventions and protocols have been formulated and implemented successfully for environmental conservation and climate change prevention. Conferences like the Stockholm Conference, 1972, the Earth Summit, 1992, the Kyoto Protocol, 1997, and many more are landmarks to show that if the countries are willing and come forward to take action, the earth can be saved. The success of all the efforts ultimately depends on how we perceive the Earth and the environment. Are we, a part of the earth, or some species that is there to consume the earth's resources. We can only answer such questions with the understanding of environmental ethics, that are the guiding principles for interactions of humans with environment.

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Important Links

Agenda 21, a comprehensive document. <https://sustainabledevelopment.un.org/content/documents/Agenda21.pdf>

International Union for Conservation of Nature, <https://www.iucn.org/about>

The Basel Convention on Transboundary Movement of Hazardous Substances, http://www.basel.int/the_convention/overview/tabid/1271/default.aspx The Convention on Biodiversity, <https://www.cbd.int/>

The Convention on International Trade of Endangered Species, <https://www.cites.org/eng/disc/what.php>

The Convention on Migratory Species, <https://www.cms.int/>

The Millennium Development Goals and indicators, <http://mdgs.un.org/unsd/mdg/Host.aspx?Content=Indicators/OfficialList.htm>.

The Montreal Protocol, <https://www.environment.gov.au/protection/ozone/montreal-protocol>

The Ramsar Convention on Wetlands, <https://www.ramsar.org/about/>.

The report of World Commission on Environment and Development, “Our Common Future”, <https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf>

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The Stockholm Declaration. https://www.ipcc.ch/apps/njlite/srex/njlite_download.php?id=6471

The Sustainable Development Goals indicators, <https://unstats.un.org/sdgs/indicators/indicators-list/>

The United Nations Convention to Combat Desertification, <https://www.unccd.int>. The United Nations Framework Convention on Climate Change (UNFCCC), <https://unfccc.int/about-us/about-the-secretariat>

7.7 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) There is a strong need for sustainable development as:
 - To fulfil the basic human requirements of food, clean water and energy.
 - The natural resources available on earth are exhausting fast and their quality is deteriorating.
 - The future generations should be able to sustain the selves on earth.
 - The economy of the world would be sustainable only when the environment
- 2) The three pillars of sustainability are:
 - Environmental viability: any progress or developmental project should include environmental conservation and protection and cause least possible disturbance to environment.
 - Economical feasibility: for ensuring long term economical growth, natural resources must be available in plenty, therefore use of resources must be wise.
 - Social Acceptability: the needs of the society should be fulfilled being mindfull of conservation.

Check Your Progress: 2

1. The Earth Summit is considered a land mark conference because:
 - It was attended by a large number of delegates from different countries, including the heads of the states.
 - It came up with a concrete action plan to protect environment - Agenda 21.
 - It also came up with treaties and protocols like Convention on Biodiversity, Convention to Combat Desertification and Earth Charter.
2. The protocol that works towards prevention of climate change is the Kyoto protocol. The protocol works towards prevention of climate change by reduction in emission of Green House Gases (GHG) like

carbon dioxide, sulphur hexafluoride, nitrous oxide, hydrofluorocarbons and perfluorocarbons. The mechanisms like emissions trading and clean development mechanisms for sharing of technology are adopted for the emission reduction in coordination with developed and developing nations.

Check Your Progress: 3

1. The environmental ethics and culture helps in protection of environment in multiple ways:
 - Ecocentric view of environment propogates conservation and protection of other species as they all have a role to play in nature.
 - Culturally we have been protecting nature-Sacred Groves.
 - Different religious practices also teaches us to conserve environment
 - e.g. “ahimsa” towards other life forms in Jainism.
2. There are three major viewpoints about environment:
 - **Anthropocentric view:** keep the needs of human beings at the core utilising natural resources.
 - **Biocentric view:** all life forms on earth are interdependent on each other and humans being most evolved have the moral responsibility to maintain the balance on earth.
 - **Eco-centric view:** all living forms on earth have a specific role to play in the maintenance of natural processes including humans who plays a role in the larger ecosystem.

UNIT 8 ENVIRONMENTALISM

Structure

- 8.0 Introduction
- 8.1 Learning Outcomes
- 8.2 History of Environmentalism
- 8.3 Ideology of Environmental Movements
- 8.4 Social Ecology
- 8.5 Deep Ecology
- 8.6 Environmental Inequality
- 8.7 Radical Ecology
- 8.8 Let Us Sum Up
- 8.9 Key words
- 8.10 Further Readings
- 8.11 Check Your Progress: Possible Answers

8.0 INTRODUCTION

The term “environmentalism” refers to an ideology, broad philosophy, and social movement which considers the concerns of environmental protection and environmental health improvement, particularly the measures which integrate the effect on animals, plants, humans and abiotic factors due to environmental changes. It is a “political and ethical movement that seeks to improve and Environmentalism, political and ethical movement that seeks to improve and protect the quality of the natural environment through changes to environmentally harmful human activities; through the adoption of forms of political, economic, and social organization that are thought to be necessary for, or at least conducive to, the benign treatment of the environment by humans; and through a reassessment of humanity’s relationship with nature. protect the quality of the natural environment through changes to environmentally harmful human activities; through the adoption of forms of political, economic, and social organization that are thought to be necessary for, or at least conducive to, the benign treatment of the environment by humans; and through a reassessment of humanity’s relationship with nature” (Brittanica online encyclopedia).

Environmentalism is basically a step to maintain equilibrium and the primary relation of humans and nature and their systems on which they are dependent to achieve sustainability. It advocates the preservation, restoration and improvement of the natural environment and critical earth system elements or processes such as the climate, control pollution or protect biodiversity. It is concerned about creating awareness regarding the importance of the environment to humans and other organisms. Continental European languages use the term “ecology” while “environmentalism” is generally used in English, having slightly different meanings (Harrison and Boy,

2018). There are many non-profit organizations which along with some legislative and corporate bodies play a role in fulfilling environmental sustainability goals.

In this unit, environmentalism is discussed with a focus on its history, objectives, ideology, and theories. We will discuss environmental inequality, social ecology, deep ecology, and radical ecology along with practice exercises which will help to attain a better understanding of the topic.

8.1 LEARNING OUTCOMES

After completing this unit, you will be able to:

- define the term environmentalism;
- discuss the ideology of environmentalism;
- explain environmental inequality; and
- discuss perspectives such as like deep ecology, social and radical ecology.

8.2 HISTORY OF ENVIRONMENTALISM

Concern for the environment has very much been a part of human history and is reflected in many religious systems which consider Nature to be sacred. The impact of air and water pollution on human life and health was noted even in the Roman times. Between the late 14th century and the mid-16th century AD, pollution was seen to be associated with the spread of epidemics. Soil conservation was practiced in China, India, and Peru as early as 2,000 years ago. However, these concerns did not lead to social movements and activism like the contemporary environmental movements. The Industrial Revolution and its impact upon the natural environment and on human health, can be considered as the key factor in the emergence of the contemporary environmental movement in the late 19th century Europe and America.

The American environmental movement began in the late 19th century with the consideration of preservationist issues. Nature was considered as something to be enjoyed and valued and not merely a factor of production. Some of the notable figures were Gifford Pinchot (1865–1946), the first chief of the U.S. Forest Service, who considered conservation as wise and efficient use of resources. John Muir (1838– 1914), founder of the Sierra Club, and Aldo Leopold (1887–1948), a Professor of wildlife management played a key role in the establishment of America's first national wilderness area in 1924- the Gila National Forest in New Mexico. John Muir, an inventor, geologist, botanist, wanderer and one of the earliest and well-known environmentalists, got into political battles to conserve the Yosemite, Kings Canyon and Sequoia areas as national parks. These efforts contributed to the early environmental movement in North America. Because of the preservationists' agenda in 1960s and the Land and the Wilderness Act and the Water Conservation Fund of 1964 and the National Trails System Acts and the Wild and Scenic Rivers Acts of 1968, the first wave of national environmental legislation started.

KEY ENVIRONMENTAL ACTS

Environmental movements prompted people to conserve and protect the earth's precious natural resources. Some key environmental acts enacted in the United States are listed below.

- 1) **1963 Clean Air Act:** provided funds for federal government research. This Act was the first environmental law enacted in the United States which included citizen suits and was designed to control air pollution.
- 2) **1964 Wilderness Act:** Conserved nine million acres of wilderness in national forest areas in US. The current number of areas designated as wilderness areas is 757.
- 3) **1965 Water Quality Act:** Objectives included maintaining and restoring the biological, chemical and physical quality of water; governing the states to address pollution and providing assistance to do so in the US.
- 4) **1968: National Scenic Trails Act, Wild and Scenic Rivers Act:** This Act designated 8 rivers for immediate introduction and other 27 to be evaluated. Pacific Crest and Appalachian trails were designated as first two national scenic trails.
- 5) **1970: National Environmental Policy Act:** NEPA was important to require Federal agencies for evaluating the environmental effects; all executive federal agencies are required to prepare environmental assessments (EAs) and environmental impact statements (EISSs).
- 6) **1970: Resource Conservation and Recovery Act:** controls the disposal of solid as well as hazardous waste in the US.
- 7) **1972: Federal Environmental Pesticide Control Act:** Started in United States which was set up to regulate pesticide and to control its applicators, consumers, and thereby environment.
- 8) **1973: Endangered Species Act:** approved to protect critically endangered species. "Consequence of economic growth and development untended by adequate concern and conservation".
- 9) **1976: Toxic Substances Control Act:** ensures that manufacturers test their products for risk to human health and the environment.
- 10) **1980: Comprehensive Environmental Response, Compensation, and Liability Act:** Formed to regulate and clean up sites which are contaminated with hazardous substances; these are called as "Superfund" sites. There are total 40,000 federal Superfund sites across the country, and 1,600 of them have been listed on the NPL list.
- 11) **1980: Alaska National Interest Lands Conservation Act:** Preserved 104 million acres of wilderness in Alaska.
- 12) **1990: Clean Air Act:** strengthened regulation of air pollution.

The modern-day multifaceted environmental movement in the United States began in 1970s. This movement originally considered few important environmental issues of that time. The best-seller "*Silent Spring*" by Rachel Carson, who came to be known as the "mother of the environmental

movement” was published in 1962 and drew attention to pesticides and their noxious effects on nature. As we have seen earlier, in the first decades of the twentieth century, environmentalism was commonly called as conservationism and differed from the modern environmental movement. Therefore, the conservationists of that era put more focus on the effective use of natural resources. Consumer activity is regarded as the origin of modern environmentalism that demanded safe, healthy and attractive environment as part of a higher living standard for managing natural resources.

Indian Environmental movements: Environmental movements also took place in the developing countries like India. The history of environmentalism in India is believed to have started in 1900's, during the British colonial period. The protests in rural communities have initiated environmentalism in India whereas in the West it was started by scientists. On the 27th of March 1973, a group of peasants in a remote Himalayan village stopped a group of loggers from felling trees. Thus was born the Chipko movement, and through it the modern Indian environmental movement was born. In the West, the desire to keep places unpolluted and beautiful initiated the green movement while in India environmentalism was necessary for survival not only for leisure. Local communities are dependent on the resources for their survival but are also used for urban and industrial interests which use them for profit. The policies incline towards the latter, which then leads to the protests for sustainability of nature. The Chipko movement in the Kumaon, the struggles of tribals of Chhotanagpur, protests by tribals of Gandhamardan (Orissa) due to damage caused to their local environment because of bauxite mining all point out to environmental degradation. When Indira Gandhi returned to power, in 1980, a Department of Environment was established at the Centre. This became full-fledged Ministry of Environment and Forests and is presently the Ministry of Environment and Forests and climate change. New laws were enacted to control pollution and to protect natural forests. In 1986, The Environment Protection Act was enacted.

At the same time, journalists had begun to analyse studying the impact of environmental degradation on social life across India. The pioneering reports were from Anil Agarwal, Darryl D' Monte, Kalpana Sharma, Usha Rai, and Nagesh Hegde. They played an important role in making the citizens aware of environmental problems. Thus, environmentalism of the poor entered educational institutions. Courses were offered on environmental sociology and environmental history. In this way, elements of an environmental consciousness had, finally, begun to take an important place in the Indian society.

8.3 IDEOLOGY OF ENVIRONMENTAL MOVEMENTS

The environmental movements also have a distinct ideology like any other social movement. An ideology can be defined as 'a group of ideas which give authenticity to an existing or proposed system of relationships, and correspondingly support an action programme to sustain the prevailing system'. It is an essential element of a social movement and is action-driven

and provides inspiration to the social movement. It also provides a framework for the movement. There are diverse actors and issues of the environmental movements and its ideologies. The environmental movements have two different ideological orientations. The first is 'ecocentric ideology' and the second is 'technocratic ideology'. The former is more radical than the latter, puts the environment first and also argues for a fundamental change in already existing social and economic uses of nature. On the other hand, the technology centric ideology is orthodox and considers that the environment can be rationally treated within the existing structure. But this ideology is not accepted universally.

An **ecocentric viewpoint** integrates social, spiritual and environmental dimensions into a holistic ideal. It puts ecology and nature as central to humanity and emphasizes a less materialistic approach to life with greater self-sufficiency of societies. An ecocentric viewpoint prioritizes bio- rights, emphasizes the importance of education and encourages self-restraint in human behavior.

An **anthropocentric viewpoint** argues that humans must sustainably manage the global system. This might be using taxes, environmental regulation and legislation.

A **technocentric viewpoint** argues that technological developments can provide solutions to environmental problems. This is a consequence of a largely optimistic view of the role humans can play in improving the lot of humanity. Scientific research is encouraged to form policies and to understand how systems can be controlled, manipulated or changed to solve resource depletion.

8.4 SOCIAL ECOLOGY

Social ecology is a school of thought which lays emphasis on small-scale economic structures and the social dimensions of the ecological crisis. Its major proponent was Murray Bookchin. "Social ecologists trace the causes of environmental degradation to the existence of unjust, hierarchical relationships in human society, which they see as endemic to the social structures of modern capitalist states. Accordingly, they argue, the most environmentally sympathetic form of political and social organization is one based on decentralized small-scale communities and systems of production" (Britannica Online Encyclopedia). The followers of social ecology consider that social relationships shape the western ideology of influencing nature. Murray Bookchin says: "The domination of nature by man stems from the very real domination of human by human". The most serious ecological dislocations include economic, ethnic, cultural and gender conflicts and many others. Social ecology believes in the interdependence of the socio-cultural domains and biophysical domains. The roots of social ecology lie in the interconnections it forms between society and ecology. Most of the present ecological issues arise from deep-seated social problems and these cannot be understood and explained without understanding of reality of our society and the impracticality within it.

Social ecology stresses direct relation of the human and non-human world, the society which is based on ecological principles; has unity in diversity, is free from hierarchical organisation and respects mutual interrelationship in every aspect of life as a suitable alternative.

It is an appeal for social reformation and moral regeneration among ecological lines. According to Murray Bookchin “Social ecologists believe that things like racism, sexism, third world exploitation are a product of the same mechanisms that cause rainforest devastation”. Bookchin has a strong opinion about the transformation of human-nature relationship if there could be change in human society. The basis of social ecology is that “ecological problems arise from deep-seated social problems and ecological problems cannot be understood, much less resolved, without facing social issues.”

According to Bookchin, the root causes of environmental problems are the market economy and the capitalist system which include trade for profit, industrial expansion, and the identification of progress with corporate self-interest

8.5 DEEP ECOLOGY

Deep ecology is defined as environmental and ecological philosophy that promotes inherent value of living beings disregarding of their utility to human requirements and also reconstruction of modern human societies in accordance with these ideas. Like the social ecologists, deep ecology distrusts capitalism and industrial technology and favours decentralized forms of social organization.

The phrase “deep ecology” was given by Arne Naess, a Norwegian philosopher, in 1972 on an isolated region of Hallingskarvet Mountain in central Norway where he thought, meditated and wrote.

Deep ecology is an environmental philosophy and social movement based in the belief that humans must radically change the way they relate to nature. Rather than valuing nature only because it is useful to human beings, deep ecology believes in the inherent value of nature. It considers humans no more important than other species and suggests readjustment of the relation of humans and nature.

Deep ecology is based on the following principles:

- **Inherent value-** The flourishing life on the earth has value in itself.
- **Diversity-** Diversity of life forms contribute to the understanding of these values.
- **Vital Needs-** Humans can fulfil their vital needs but have no right to deplete this diversity.
- **Population-** The flourishing of human life and cultures is compatible with a substantial decrease of the human population. The flourishing of nonhuman life requires such a decrease.
- **Human Interference-** The situation is rapidly worsening due to huge human interference with the nonhuman world.

- **Policy Change-** The basic economic, ideological and technological and structural policies are required to be changed.
- **Quality of Life-** The direction of change should be towards increasing life quality instead of increasing higher standards of living.
- **Obligation of Action-** There is an obligation to implement the essential changes.

Deep ecology has been criticised for not giving due importance to human life. According to Bookchin, deep ecology wrongly identifies human beings and their activities as the major cause of environmental problems. He compared deep ecologists to radical environmentalists who favour anti-human population controls.

Difference between Shallow and Deep Ecology

The core belief of deep ecology is that all the life present has the equal right to live on the earth. This refers to the consideration that the interests of other living beings should also be valued as that of humans. For example, a rainforest can no longer be treated as a valueless source of wood only instead, a collection of living creatures who all have equal right to live and flourish. Nature is said to have intrinsic value. It is valuable even if humans can't find use from it. From a deep ecologist's point, protecting humans from climate change will have effect on other living beings and they would suffer. Deep ecology says that we should expand our view of who we are so that it also considers the natural world. This is sometimes regarded as the expanded self. We are harming ourselves, if we harm nature. Deep ecology favours ecocentrism or biocentrism and rejects anthropocentrism. On the other hand, Shallow ecology rejects ecocentrism and biocentrism. They consider that there is nothing wrong with the anthropocentric view. Nature is there to serve human interests. This is sometimes called as instrumental value. Also, climate change is bad because human interests will suffer from this. It is humans that will ultimately suffer if climate change is allowed to occur. Protecting humans from the effects of climate change is still considered a bad thing as the damage caused to other life forms would also affect humans. It will lead to difficulty in obtaining natural resources. It treats richness and diversity of species as intrinsic values are based on respect for nonhuman life forms. Humans are considered part of nature instead its ruler. On the other hand, shallow ecology fights resource depletion and the pollution for the sake of the health and wealth of people in the developed countries. It is more dependent on the efficient use of the environment rather than to work for the environment. Naess refers to it as management ethics instead of environmental ethics and thus highlights its selfish and exploitative nature and social injustice. The short term policies of shallow ecology are not fair because they ignore the underdeveloped countries. Shallow Ecology is supported by Anthony Weston, an American writer, teacher, and philosopher. Weston's philosophical project as a whole advances an expansive "toolbox" for critical, creative, and constructive thinking, especially for purposes of social and environmental re-imagination and pragmatic ethical practice.

In India environmental movements are characterized by free-floating, eclectic brands of ideologies. Ramachandra Guha is an Indian scholar who has

contributed to the understanding of 'environmentalism in India' and discussed the various strands of environmental thinking which are detailed below. Control over natural resources, socioeconomic reasons, environmental destruction and wide spread awareness and media can be regarded as the reasons of the origination of environmental movements in India.

'The Crusading Gandhians'

The Crusading Gandhians found strong adherents in England and Germany because industrialization brought a comeback of agrarian sentiment. India during Pre-industrialized time contributed a more practical agrarian thinker in Mahatma Gandhi, who read Ruskin and Carpenter while studying in England. Scientific conservation with a concern for environmental degradation, also took root in Germany and Britain. Different environmental movements across the globe areas and different environmental philosophers use the ideas of Gandhi like green and deep ecology etc. Petra Kelly, founder of German green party was influenced by Gandhi and wrote "a lifestyle and a method of production which rely on an endless supply of raw materials and a lavish use of these raw materials generate the motive for the violent appropriation of raw materials from other countries". Arne Naess known as father of deep ecology was influenced by Mahatma's metaphysics. He accepted his work on ecology which was developed from the work of Spinoza and Gandhi. Naess defines bio-spherical egalitarianism as the internal relation between self-realization and non-violence given by Gandhi. He gives credit to Gandhi for his work on Deep Ecology which helped him to become champion of environmentalism.

The Ecological Marxists

They follow Gandhian tradition as a mode of action like padayatras, fasts and poojas. Guha says that the social and ecological harmony is the result of the pre-capitalist and pre-colonial community according to crusading Gandhians. The believers of appropriate technologies tried for agriculture and industry and technological methods. While the ecological Marxists do not believe in traditions and focus only on the scientific facts. *Kerala Sastra Sahitya Parishad (KSSP)* is an example of ecological Marxism, according to Guha.

A study of the Silent Valley movement shows the practices of Mahatma. Gandhian practices were adopted in the movement against the environmental injustice. The followers belong to the different areas of society like journalists, social workers, intellectuals, students, and teachers etc. They raised the awareness to the people about the role of the Silent Valley Forest to the southern monsoon. Modern industrialization is considered the root cause of environmental deterioration according to Gandhi whereas Marxists believe capitalism as its major cause making both the ideologies different. According to Gandhi, human needs are to be limited for the sake of nature while Marx considers every man's needs. Marx considers science and technology as a tool for regulating nature, on the other hand, Gandhi sees it as an obstacle to nature conservation.

Scientific Conservationists

Scientific conservationists focus on the efficiency and management whereas wilderness enthusiasts are concerned for their commitment about the protection of biodiversity. Here is the notable work of Scientific Conservationists B.B. Vohra. He was one of the first scientific conservationists to draw public attention to land and water degradation. In 1973 he documented water erosion, water-logging and other types of land degradation. In 1973 he documented the extent of water erosion, water-logging and other types of land degradation. No coordination between concerned departments of the government was there and national organisation or policies were formed to combat with these problems. Formation of new ministries and departments were the only solution for him at that time.

The Appropriate Technologists

Appropriate Technology is the most practice oriented among all other ideologies which focuses to develop and apply technical and institutional options to the centralizing and environmentally destructive process of development in India. The appropriate technologists think in the amalgamation of between eastern and western traditions and decentralized planning and adherence to small scale non-destructive economic activities.

Thus, appropriate technologists have worked for the generation and diffusion of resource conservation, socially liberating technologies and labour intensive. Demonstration of the practices as a set of socio-technical options to the presently operating technologies, but, not so much on challenging the 'system'.

Sarvodayee Chandi Prasad Bhatt guided the Chipko Movement which is basically Gandhian in its ideology, but this movement has also accepted the principles of appropriate technology as commended by Bahuguna. Narmada campaign is the interaction of Gandhian ideas, appropriate technology, and politico economic frame in the Marxian ideology.

Wilderness Enthusiasts

All the three Crusading Gandhians, appropriate technologists and ecological Marxists represent the three most powerful ideologies in the Indian environmental movement but the other two ideologies i.e. scientific conservationists and the Wilderness Enthusiasts also have a significant support. Indian naturalists have provided huge evidence for the decline of natural forests and the flora and the fauna species, appealing the government to take restorative action. Both scientific conservation and wilderness ideologies have considerable influence on government policies rather to command popular following.

Two more ideologies were talked about by Baviskar in 1995 - the Ideology of Conservation and the ideology of Indigenous Ecological Management. Ecofeminism is also one of the ideologies of Indian environmental movements. Baviskar emphasizes on the 'environmentalism of the poor' and says the poor people are mostly dependent on natural resources or on

environment to support their livelihood and existence; any form of environmental degradation will majorly affect their life and survival. She verified the argument of 'full-stomach' environmentalism and 'empty belly' environmentalism. She connects the link of environmentalism between Southern movements and the Northern practices. Northern practices have strengthened the Southern movements and helped them to develop the idea of 'red' and 'green' nature of Indian environmentalism. Study on Industrial Pollution in Delhi and Fisheries in Kerala by Kavita Philip, Amita Baviskar and Subir Sinha developed the interrelationship between two types of environmentalism i.e. environmentalism of politically powerful class and the poor working class of factories and fish workers. These two groups focus on the equal distribution of resources and safe working conditions. They have raised anti-pollution movement and fish workers movement.

The ideologies are different and often do not overlap. A single environmental organization or group may follow more than one ideology. On the other hand, a single ideology may motivate many environmental movements at a given time. The procedure of formation and denial of the ideologies of such movements in the past have slowly given rise to a new ideological form that is 'Ecologism' which is its budding stage.

Check Your Progress: 1

Notes: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What do you understand by environmentalism?

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2) What are the objectives of Environmentalism?

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8.6 ENVIRONMENTAL INEQUALITY

This refers to how environmental quality differs between different communities of the country in several features like air pollution, availability of green space and exposure to flood risk. Environmental inequality has detrimental effects on individuals as well as communities. The following important aspects must be kept in mind to ensure equity:

- i) ***Intrinsic value of environmental equity:*** The fundamentals of legal documents and constitutions consider that everyone has a right to safe and clean environment. These values do not apply for individuals but also distributed across communities. A more equal distribution of the effects of the exposure to pollution can be considered as ethically correct and preferable to unequal distribution.

- ii) ***Equality of opportunities:*** Equality of opportunity is the second concern. According to the World Bank's Human Opportunity Index, "Much more important than inequality of outcomes among adults is inequality of opportunity among children". The important point of the discussion should not be about equality but about equity which means equal chances for all but not equal rewards for all.
- iii) ***Economic impacts:*** Pollution also affects in terms of economic impacts on property, work loss and health issues. According to the EPA (2014) effects are "especially important to minority and low-income populations who are disproportionately impacted by asthma and other debilitating health conditions." Environmental justice takes care of enactment of environmental policies and laws so that no group or community shares a disproportionate ratio of harmful impacts of environmental hazards and pollution, eg. air and water pollution, inadequate access to healthy food, unsafe homes, and inadequate transportation etc. Some communities or human groups have a disproportionate share of higher levels of environmental risk than other sections of society and this has initiated the concept of environmental injustice and inequality.

To give an example from India, we may consider how a small but increasing number of people drive a car. In Delhi, 15 percent of the people drive. Their driving cost is high in terms of air pollution. The question arises, how to mitigate air pollution because more and more people in Delhi and other car-clogged cities drive. Is it possible for the remaining 85 per cent to plan? Is there enough space on the road or in the already polluted 'air shed' for the cars of the many? The answer is clearly, no. Unless we restart urban mobility on an uncommon scale, we cannot achieve clean air. Understanding this, the Delhi High Court considering 'equity of use' in 2010 ruled that Delhi roads need to be planned. Today, the bulk of Delhi citizens walk, cycle, or take a public transport bus or metro train. Most of the poor people do this. The question here is whether these cities having poor-rich gradation of the evolving world can be restarted upgraded and be improved so that the urban public transport systems is used more, and the majority won't need to use private vehicles any more. This will help cities to not become car-clogged and then try to accommodate public vehicles. The approach to make policy to mitigate air pollution must move from cleaning the tailpipe to planning for affordable and overall mobility solutions. The solutions must work for the poor and for the rich.

To overcome the issue of the environmental injustice solutions should be given rather than just posing problems. The technologies and approaches for the management of environmental problems will have to be recognised differently so that sustainable growth is achieved. At present, a development that ensures the sustainable future for upcoming generations is the only way that will work on ground level to combat all sorts of environmental issues and inequalities. This puts a new way towards environmentalism to adopt ideas only with idealism and purpose.

8.7 RADICAL ECOLOGY

Radical ecology focuses on the solutions to environmental crisis which are more beneficial than market incentives or technological fixes. Radical ecology is an evaluative approach that works through thought and praxis, and focuses on how we can really resist the destructive process of industrial growth and socio-economic system so as to bring the changes which helps to achieve new strategy for living which follows the natural world. It involves the work and experiments in new economics, fixed ideological position, process-oriented science, revitalised spirituality and hierarchical social forms.

It is thus not a unified stream of thought that works from or towards a fixed ideology. It is a way of thinking that searches within a certain 'bandwidth' how a radical transformation of human 'being in the world' can be brought about, that would allow humans and non-human beings both to flourish. From the perspective of Radical Ecology such flourishing can only happen if we focus on the ecological or interrelated nature of life, without losing sight of the particular needs of all species. In that sense it is a form of Deep Ecology that is not satisfied with finding (shallow) legal, technical and institutional solutions to the ecological crisis. Radical Ecology stresses the need for a change in consciousness, but, also for the need of a transformation of humanity's concrete ways of acting and understanding itself. It is here that the concept of ahimsa as it is understood in nonviolence thinking might have something to offer.

Check Your Progress: 2

Notes: 1) Use the space provided below for your answers.
2) Compare your answers with those given at the end of the unit.

1) What is Environmental inequality?

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8.8 LET US SUM UP

In this unit we have described the term environmentalism and its history, ideologies, and theories. We know that the environment of the Earth is changing due to various anthropogenic activities which lead to unsustainability. This unit focuses on environmental aspect of green ideology and politics and social ecology. The requirement of environmentalism is basically a step to equilibrium to maintain the primary relation of humans and nature and their systems on which they are dependent to achieve sustainability. It is mainly concerned about creating awareness regarding the importance of the environment to humans and other organisms.

8.9 KEY WORDS

Environmentalism: The term “environmentalism” is considered as an ideology, broad philosophy, and social movement which takes into account the concerns of environmental protection and environmental health improvement, particularly the measures which integrate the effect on animals, plants, humans and abiotic factors due to environmental changes.

Ideology: An ideology can be defined as a group of ideas which give authenticity to an existing or proposed system of relationships, and correspondingly support an action programme to sustain the prevailing system.

Deep ecology: *Deep ecology* is define as environmental and ecological philosophy that promotes inherent value of living beings and also reconstruction of modern human societies in accordance with these ideas.

Social ecology: Social ecology is the hierarchical arrangement of authoritarian mentality and power that has been planted in the society.

Shallow ecology: It rejects ecocentrism and biocentrism. They consider that there is nothing wrong with the anthropocentric view. Nature is there to serve human interests.

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8.11 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) The term “environmentalism” is considered as an ideology, broad philosophy, and social movement which takes into account the concerns of environmental protection and environmental health improvement, particularly the measures which integrate the effect on animals, plants, humans and abiotic factors due to environmental changes. It is a theory which focuses on environment first than heredity as a crucial factor of the evolution, especially cultural and intellectual development of an individual or group.

2) Objectives of Environmentalism are:

- To change how people interact with the environment and natural resources.
- Form public policies to protect the environment.
- Create awareness about how people affect the environment through different ways.

Check Your Progress: 2

- 1) The environmental quality which differs between different communities of the country in a number of features like air pollution, availability of green space and exposure to flood risk.

The main objective of deep ecology is the protection of non-industrial cultures from the invasion of industrial societies. Deep ecology focuses on the biological richness and diversity of life forms.



UNIT 9 SOCIAL AND CIVIL SOCIETY MOVEMENTS IN INDIA

Structure

- 9.0 Introduction
- 9.1 Learning Outcomes
- 9.2 Early Environmental Movement in India: The Bishnoi Movement in Medieval Western Rajasthan
- 9.3 Bishnoi Philosophy of Conservation of Vegetation and Movement for Environmental Protection
- 9.4 Colonial India: Forest Policies, Tribal Land Displacement and Deprivation of Livelihood of the Adivasis
- 9.5 Routine Revolts by Tribes of Forests
- 9.6 Forest Act of 1865, 1878, 1972 and 1935. The Case-study of Bastar in Pre-independence and Post-Policing of Forest Land and Increasing Rural Taxation
- 9.7 Van Mahotsava in Independent India: The National Forest Policies of 1952 and 1988
- 9.8 Environmental Movements in Contemporary India: Chipko Movement, Narmada Bachao Andolan and Appiko
- 9.9 Let Us Sum Up
- 9.10 Key Words
- 9.11 Further Readings
- 9.12 Check Your Progress: Possible Answers

9.0 INTRODUCTION

Archaeological history and anthropological evidences clearly indicate that human civilization shaped the contours of natural environment throughout the centuries. From pre-historic times to the colonial times, society and state shaped the contours of natural environment for the economic growth and sustainability of a growing India civilization and state-societies. In this introduction, here we are not arguing against or in favor of agricultural expansion versus environmental movements. Here is a brief historical narrative of the causative factors for the emergence of social and civil movements for conservation of natural environment.

9.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- define the social and civil society movements in Environment;
- describe a brief history of movement for conservation of natural environment; and

- explain about environmental activities in contemporary India.

9.2 EARLY ENVIRONMENTAL MOVEMENT IN INDIA: THE BISHNOI MOVEMENT IN MEDIEVAL WESTERN RAJASTHAN

The human society eked out their living in the pre-historic times (Paleolithic and Mesolithic period) through hunting and gathering. Neolithic age (new Stone Age) witnessed the beginnings of subsistence agricultural activities and emergence of villages. As subsistence agriculture expanded into surplus agriculture, we witnessed the beginnings of urbanization and emergence of cities. The first four river valley civilizations of the world are Indus valley civilization in South Asia (India and Pakistan), Nile Valley civilization in Egypt, Hwang-Ho valley civilization in China and Euphrates Tigris valley civilization in Iraq. The urban society utilized water-resources for agrarian expansion to produce surplus production to sustain majority of the non-agricultural, urbanized section society. It is important to know that it is the environmental factors which were responsible for the decline of Indus/Harappa urbanization (Cradle of Indian and South Asian Civilization) such as earthquake, floods and silting of river Indus and its major tributaries. It is obvious that over-utilization of natural resources without ensuing regeneration played havoc with Indus valley civilization and was responsible for its decline.

In Mauryan empire is known for tapping into natural resources in strengthening its revenue base. A detailed study of Asokan inscriptions and Kautilya's Arthashastra clearly reveal that the Mauryan state expanded agricultural activities into forest and tribal regions to augment their revenue income. Heavy taxation system of the Mauryan encouraged fallow land to be brought under cultivation and peasants were exempted from agricultural taxes for bringing fallow land under cultivation. It is equally interesting that the ancient world was concerned about environmental protection such as the measures taken by emperor Asoka in conserving natural resources through afforestation plantation, digging of wells, ban on the killing of deer's and peacocks in royal kitchen on certain days. Early medieval period of Indian History (AD 7th –AD 13th Centuries) witnessed processes of state formation and agricultural expansion further into hinterland and tribal habitat throughout the sub-continent. It is true that plenty of availability of land did not lead to any social or civil movement as we do not have any documented evidence of environmental movement for early India. However medieval India witnessed conflicts just like medieval Europe over common resources. It is in the Thar Desert of India that the first civil movement for environmental protection arose which has been well documented in the historical literature. This is the famous Bishnoi movement of western Rajasthan founded by Saint Jambhoji in Bikaner in 1453.

Bishnoi Movement in Medieval Rajasthan

One of the earliest environmental movements in historic India was the Bishnoi movement in medieval Rajasthan. It was founded by a Parmar Rajput, Jambhoji, Bikaner. Jambhoji was born to a Parmar Rajput named

Lohat and his Bhati Rajput wife Hansa. An agro-pastoralist, Lohat received divine message (akashvani) that a Krishna-like avatar would be born to him to liberate 12 crore human beings. It was forecast that this son would be a yogi (sage). Hansa gave birth to Krishna like Jambhoji. In fact, Jambhoji, like infant Krishna performed miracles and surprised visitors to his home. He like Krishna was born with an aura. Like Vishnu, he (Jambhoji) lit lamps with water and not oil. Hence, Vilhoji adapted Krishna's childhood deeds to a significant extent to highlight the avatar status of Jambhoji. Vilhoji repeats that Jambhoji is the Vishnu re-incarnate in Kali Yuga. All the Vishnoi saints, including Alamji, legitimized Jambhoji's avatar and divine status through appropriation of Vishnu's and Krishna's characteristic. Vilhoji makes another interesting identification of Jambhoji, with Lakshmana of Ramayana who killed Ravana and rescued Sita from Lanka.

9.3 BISHNOI PHILOSOPHY OF CONSERVATION OF VEGETATION AND MOVEMENT FOR ENVIRONMENTAL PROTECTION

The Bishnoi movement had a wide following in the desert community of western Rajasthan in the medieval period. Its philosophical and socio-economic content enjoyed popularity among the local population of Rajputs, Jats and Rabaris of Raikas (the caste of camel breeders) because it made a significant difference to their life. The Bishnoi concern for environmental protection ensured improvement in the living conditions of the desert community. This concern assumed significant proportion against the background of the famines of AD 1450, 1485 and 1490 in western Rajasthan followed by those of 1742, 1747, 1792, 1796, 1804, 1812-16, 1837-8, 1866-9, 1890-2, 1898-1900, 1901-2 and 1938-9. Bikaner region witnessed a series of famines in 1834, 1849, 1860, 1868-9, 1889-92 and 1899. The fact that the Bishnois of Rajasthan and Haryana are a living culture today, despite the recurrent famines indicates the continuing relevance and contemporarily of the philosophy and teaching of the medieval saints (Nandini Sinha Kapur, 'Tribes, Agro-pastoralists and Environment in Western India ed. 7th -20 Centuries, New Delhi. 2011).

Following Jambhoji's preaching's, Vilhoji widened the scope for conservation of nature by highlighting the environmental and economic importance of vegetation, particularly the Khejri tree. The Bishnois were not only inspired to sacrifice their lives for the protection of cattle but also for the preservation of trees. Vilhoji warned his disciples that they would go to hell if they did not demonstrate kindness towards trees. He instructed his disciples to plan the right kind of trees which would bear nectar-like and not poison-like fruits. He compared truth and falsity with two types of trees: sweet and salty. Vilhoji narrated the story of Jambhoji's disciple Karamni who sacrificed her life in 1604 for the protection of Khejri trees – the lifeline of the desert as the trees can survive severe drought conditions. Khinvani and Netu are two other heroes who laid down their lives for saving trees.

Their accounts assert that if trees are not protected then gurudharma (teaching of Jambhoji) would be disrespected. Bhati Rajput are condemned

for felling trees and other Jambhoji follower Mote gave up his life fighting against these Bhatīs. Glorification of these martyrs is sought as hero Netu attains heaven and immortality. Jambhoji's seat Sambharthal is hailed because it abounds in forests and is habited by devout pastoralists. Jambhoji wandered in these forests but never cut any tree for wood. He only picked up broken twigs to light fires. Jambhoji encouraged his followers to grow mangoes out of thorny desert oaks. Among his miracles, the story of conversion of lime trees into coconut trees continues to be popular among his disciples. When human beings and animals seek shelter in the shadows of trees, they get respite from heat. Interestingly, Vilhoji laments that trees shed their leaves as soon as they are surrounded by human being and animals. This is a clear indication of the Bishnoi preference for conservation of vegetation.

Water Resources and Agriculture

Vilhoji popularized conservation of the other scarce item of the desert: water. Conservation of water was not only to meet the daily needs of a desert society but also for agricultural activities. According to Vilhoji, when rain comes, cows get green grass to eat and water to drink. Hence, water is the foundation for good quality of milk and not any hymn or mantra. It was imperative to conserve rain water at the local level both for drinking purposes and pastoral activities. Vilhoji prayed to Devji (Jambhoji) to bestow kindness on earth by sending rain-bearing clouds.

It is significant that the need for water and water-conservation were sanctified though such comparisons as 'Water can exist without fish but fish cannot survive without water, just like guru Jambhoji does not need his disciples as much as his disciples need him'. When it rains, cattle grow up healthy and give plenty of milk. Milch cow graze on green grass that only grows after rains. Good quality fodder like bagarau and chandlevo grass grows only if it rains. To sanctify the scarcest resource of the desert, water is claimed to be the sacred Banganga of the Puranic traditions.

The need for generation and conservation of water perhaps was the most important factor in the cultivation of grains for the sustenance of the local population. The Bishnoi concern for agricultural activities to meet the local requirements is evident in Vilhoji's narrative of the famine of 1542 in Katha Guggalai ki. The famine was so terrible that the starved population began to migrate elsewhere in search of livelihood. When Jambhoji attempted to stop the people from migrating, they petitioned that if their daily requirement of 1.25 maunds of grain were arranged for, they would not migrate in search of livelihood. Jambhoji promised them daily their food through divine intervention. But in lieu he demanded that the local population promise to save animals and birds and not kill them. If they promised to protect the animal-wealth only then God would remove their difficulties. This is a Bishnoi principle that connects conservation of cattle wealth with agricultural wealth. People doubted Jambhoji's claim of divine generation of food. But Vilhoji writes that wherever Jambhoji preached, those places produced more grains, fruits and flowers. In another account Jambhoji is said to have gone a forest accompanied by a Purbia and his camel laden with sacks. The Purbia was delighted to see maunds of grains which Jambhoji procured from the

forest. The starved population acknowledged this miracle and received grain from Jambhoji every alternate day.

The need for a desert economy to trade in grains with its neighboring states is also a Bishnoi concern. As severe winter gave way to summer, the local people prepared their camels to leave for trading. They visited Sindh to buy wheat and came back to plough the fields before the monsoon. In another incident, Jambhoji created a camel out of guggal (sticky material meant to be burnt as incense) and loaded it with sacks of grain and sent Purbia with money and camel to the villages of Bishnois. Hence, Jambhoji fed famine-struck people with grains and emphasized upon generation of grains through local cultivation and trade with fertile regions like Sindh. Water and good quality agricultural production are inextricably linked with each other and are always reflected in Vilhoji's teachings. Trees which receive plenty of water grow to have large branches and bear fruits and render cool shadows. Those who look after the trees reap the fruits and hence, we should respect trees. Fertility of soil and seeds are sanctified by comparing these with a knowledgeable guru who can liberate a disciple. Vilhoji compares chanting of God's name with continuous irrigation. He teaches his disciples to fill wells water and irrigate plants just like they should chant God's name till the end, then gardens would be laden with fruits and flowers.

If agriculture was essential to sustain and sedentarize a desert society and to meet the increasing revenue demands of the Rathaur state of Jodhpur and Bikaner, then pastoral activities was the lifeline of the desert economy. The Bishnois not only uphold preservation of cattle wealth but also relate cattle and dairy farming to the worship of their chief deity Vishnu. Vilhoji instructs his disciples to strain the water before they drink to save micro-organisms. He condemns those who eat meat instead of good food (meaning vegetarian food) in the same context. For the same reasons, Vilhoji warned against lighting fire without regard to micro-organisms; if a fire burns organisms, then it becomes inauspicious. He also encouraged people to walk less to minimize the killing of small organisms.

In Katha Avatarpat (the story of Jambhoji's birth), the predominance of pastoral wealth over agricultural activities is clearly evident. Jambhoji's father, Paramar Lohat, who cultivated his field and irrigated in with water drawn from a well, had plenty of cows and goats. His agro-pastoral activities are vividly described by Vilhoji. Lohat grazed his cattle and spent money on his animals while procuring grains during the monsoon. It is when he had gone to the forests to graze his cows that the divine Vishnu appeared before him and foretold the birth of avatar Devji (Jambhoji).

9.4 COLONIAL INDIA: FOREST POLICIES, TRIBAL LAND DISPLACEMENT AND DEPRIVATION OF LIVELIHOOD OF THE ADIVASIS

Colonial rule was the first turning point in Indian history that a government claimed a direct proprietary right over forests. This was not the case in pre-

colonial times including Mughal India. The British state became the conservator of forests when it passed the Indian Forest Act of 1878. Hundreds of thousands of acres of forest lands that adivasis had used for centuries and were considered commons suddenly kept in reserve, a practice that continued for the rest of the colonial period. With British control of the forests came the concomitant rise of moneylenders, traders, and immigrants, and the influx of these new intermediary groups led to widespread adivasi land displacement and were deprived of livelihood. These are only some of the major changes instituted during the colonial period. The introduction of money rather than a barter economy—also transformed the nature of tribal society during the course of British rule.

9.5 ROUTINE REVOLTS BY TRIBES OF FORESTS

Consequently, revolts among the indigenous population became a routine phenomenon during colonialism, especially in the nineteenth century. For instance, in 1855 the Santhals rebelled; in 1868 the Naikdas; in 1873 the Kolis; and in 1895 the Birsas. This is only a small smattering of the total number of conflicts. Ranajit Guha has documented over 110 different colonial-era peasant revolts and Kathleen Gough records at least 77 since the advent of British rule.

British colonial rule in India ushered in a period of intense rebellion among the country's indigenous groups. Most tribal conflicts occurred in the British provinces, and many historians have documented how colonial policies gave rise to widespread rural unrest and violence. In the post-independence period, many of the colonial-era policies continued to plague the system and were not reformed. Tribal conflicts continued in the form of the Naxalite insurgency. The princely state of Bastar continues to be a major centre of tribal conflict in India.

Scholars have pointed out that the new colonial policies, such as the policing of forest lands and increased rural taxation, led to widespread discontent and rebellion among indigenous groups. To quote Stokes 'resentment against (moneylenders) boiled over most readily into violence among tribal people like the Bhils, Santals, and the Gonds'. Historians have shown that in post-independence, era the new Indian government did not reform a number of colonial-period policies, especially those dealing with forestry, and tribal conflicts continued to occur throughout the country, especially in former areas of direct British rule like Bengal, Bihar, and Jharkhand. The Naxalite movement became the manifestation of tribal revolt in contemporary India.

Tribal revolts began in Bastar precisely because of increasing British influence in the state. Three specific policies were implemented in Bastar that gave risotto tribal revolt: colonial officials took directly controlled the forests, they displaced tribals from their land, and they interfered into succession to the local royal throne, which upset the native population.

Bastar led to the rise of the contemporary Naxalite insurgency. The case of Bastar, reaffirms the role of British colonialism in producing tribal conflict in

India by showcasing its effects in areas that never formally came under the ambit of direct rule. Importantly, however, the continuing violence in Bastar demonstrates implicates the post-colonial government's failures to end the root causes of the bloodshed.

Despite its remote location, the political and economic developments in colonial Bastar led to persistent rebellion and provides important insights for other regions throughout Asia. The British practice of retaining areas of indirect rule within a colony was adopted from India and exported to other colonial territories such as Burma and Malaya. Therefore, understanding contemporary violence in other post-colonial states in Asia-ethnic separatism throughout former areas of indirect rule in Myanmar, for example-can be understood by analysing what happened in Bastar.

This part of this chapter contains four major sections. In the first two the general history of tribal revolts in colonial and then post-colonial India are discussed. In the final two, broad trends within the kingdom of Bastar, again in the colonial and post-colonial periods are listed.

9.6 FOREST ACT OF 1865, 1878, 1972 AND 1935. THE CASE-STUDY OF BASTAR IN PRE- INDEPENDENCE AND POST-POLICING OF FOREST LAND AND INCREASING RURAL TAXATION

The survey and mapping of India's forests allowed the implementation of scientific management. The dominant paradigm of scientific management was to pursue the maximum sustainable production, and management practices were organised around this principle. In deforested areas, commercially valuable species were planted (Sagreiya, 1967), while in some cases, mixed forests were replaced with marketable monocultures continuing with the interests of the colonial rulers, forest management, and its restrictions of access to local communities, resulted in, as described by Sagreiya (1967), 'a steady build up of forest capital.' The forest capital was depleted during World War One, but regenerated through intensive management, but again deforestation occurred during World War Two, this time far beyond its sustained production (Sagreiya, 1967). An increase of 65% over the war period does not indicate the timber not accounted for, which, by all accounts is considerably, greater, when timber procured from other sources is also considered (Gadgil and Guha, 1992). Gadgil and Guha also point out that as the war proceeds, the area covered by working plans diminishes, indicating an increase in cutting of forests from areas not covered with working plans, which would have been unaccountable.

According to Gadgil and Guha (1995), the large-scale annexation of Indian forests by the colonial state constituted a critical turning point, politically, socially and ecologically; politically, because the colonial claim to the forests represented an unprecedented expansion of state power and intervention, with a simultaneous reduction in local communities' rights. Socially, the traditional patterns of resource use were disrupted by the restrictions to local

access, and ecologically, as the forests were undergoing a process of 'commodification' which transformed their nature.

State control, which was a critical feature of the 1878 Forest Act, also facilitated the development of scientific forestry. Dietrich Brandis was the first Inspector General of Forests, and is considered the founder of modern forestry; indeed, he is often paid tribute as the 'Father of Indian Forestry' (Chaturvedi, 1998). Brandis was a botanist from Bonn University, and Germany was considered the leading European country in forest management in colonial times. One of the first tasks undertaken by the newly-formed Forest Department was to survey and map the forests of India. Forests were demarcated so that working plans could be formulated (Sagreiya, 1967).

According to the 1878 Forest Act, three types of forest were to be designated; Reserved, protected, and village. Reserved forests were supposed to be the most commercially valuable and amenable to sustained exploitation. Overall state control of reserved forests was sought, which meant relinquishment, or transferral of other claims and rights, limited access was granted. Legally, channels to contest the reservation of forests existed, though rural communities had little experience with legal procedures. Protected forests were similarly state controlled, but some concessions were granted, conditional to the reservation of commercial tree species, when they became valuable. Protected forests could also be closed to fuel wood collection and grazing, whenever it was considered necessary. As timber demand for empire increased, the state had granted limited control itself to be inadequate, thus many protected forests were re-designated as reserved forests. The act also provided for a third designation of forests in its constitution, village forests, which according to Gadgil and Guha (1992), was not under the control of the colonial government over most of India. The area of forest appropriated by the state in 1878 was 14,000 square miles, which had increased to 81,400 and 3300 square miles, for reserved and protected forests respectively, by 1900 (Stebbing, 1922). In east era India, state appropriation of forest land led to the dispossession of adivasi communities' ancestral land. In the Singbhum District of Bihar, large- scale encroachment by the Forest Department in the late nineteenth century displaced the Ho tribe from their villages in an attempt to demarcate a reserve forest. The reservation was criticized leading to conflict between the Ho and Forest Department which escalated into a 'tree war', one that still periodically erupts (Corbridge and Jewitt, 1997).

The Forest Acts, passed by the colonial administration in 1865 (Indian Forest Act), 1927 and 1935 resulted in military, strategic and administrative control of tribal and forests criminal tribe Act of 1871 was yet another important step towards absolute control of tribal habitat and their resources.

Gandhi Ji situated his views of conservation of natural environmental in the context of urbanization, factor system and nebulous industrialization. For the ideal India lived in an ideal village in natural setting. India witnessed two national forest policies in post-independent India: The National Forest Policy of 1952 and the National Forest Policy of 1988.

9.7 VAN MAHOTSAVA IS INDEPENDENT INDIA: THE NATIONAL FOREST POLICIES OF 1952 AND 1988

The Government inaugurated Van Mahotsava in India. A central Board of Forestry was set Target was to raise the area under forest to 100 million hectares. Hence, there was a focuses reduction of the long range development for forest resources and meet the growing demand of timber and fire wood. This policy was evaluated by the National Commission on Agriculture. The National Forest Policy of 1988 replaces the forest policy of 1952. The Forest policy of 1988 was concerned with protection, conservation and development of forests at the world stage, several events of recent history between in 1950s, 1960s and 1970s illustrated the magnitude of environmental damage caused by humans such as hydrogen bomb test at Bikini Alok, the infamous oils of 1967 and 1969, mercury poisoning on the people of Minamata. In 1972, the United Nations Conference on the Human Environment was held in Stockholder and for the first time united the representatives' of multiple governments. This conference led to the creation of government environmental agencies and the United National Environment program activists of anti nuclear movement and public awareness gradually gone birth to 'Green Politics) and 'environmental activism'. The first earth day was celebrated on 22nd April, 1970.

India witnessed a large number of environmental movements with conservation at its centre some of the notable movements that we are going to read in this unit are Chipko movement led by Sundarlal Baluguwa in the 1970s in Tehri who spearheaded anti Tehri Dam movement in the 1980s Appico movement in Karnataka, Narmada Bachoo Andolon led by Dr. Medha Patekar among other.

9.8 ENVIRONMENTAL MOVEMENTS IN CONTEMPORARY INDIA: CHIPKO MOVEMENT, NARMADA BACHAO ANDOLAN AND APPIKO

Sunderlal Bahuguna is a well know environmentalist, Chipko movement and a follower of Mahatma Gandhi. He believes in the philosophy of Non-violence and Satyagraha. He fought for long for the preservation of forests in the Himalayas, first as a member of the Chipko movement in the 1970s, and later led the Anti-Tehri Dam movement starting 1980s, to early 2004. He was one of the pioneering environmentalists of India, and later he and his people associated with the Chipko movement started taking up environmental issues, including large-scale protest against construction of large dams.

He was awarded the Padma Vibhushan, India's second highest civilian honour, on 26 January 2009.

Sunderlal Bahuguna was born in village Maroda near Tehri, Uttarakhand on 9 January 1927. He claimed in a function arranged at Kolkata, that his ancestors bearing surname Bandopadhyaya, migrated from Bengal to Tehri,

some 800 years ago. Early on, he fought against untouchability and later started organising hill women in his anti-liquor drive from 1965 to 1970. He started social activities at the age of thirteen, under the guidance of Shri Dev Suman, who was a nationalist spreading message of non-violence and he was with Congress party of Uttar Pradesh (India) at the time of Independence. Bahuguna also mobilised people against colonial rule before 1947. He adopted Gandhian principles in his life and married his wife Vimla with the condition that they would live among rural people and establish ashram in village. Inspired by Gandhi, he walked through Himalayan forests and hills, covering more than 4,700 kilometres by foot and observed the damage done by mega developmental projects on fragile eco-system of Himalaya and subsequent degradation of social life in villages. (Wikipedia, Sunderlal Bahuguna).

Chipko Movement

Chipko movement was started in 1973 spontaneously in (Uttar Pradesh) to save trees and forests from felling by forest contractors. In Hindi, “Chipko” literally means “stick” and people started hugging to trees when it was being cut. Chipko movement later inspired Appiko Movement in Karnataka. This was one of Sunderlal Bahuguna’s lifelong contributions to the cause, of environment. He created of the Chipko’s slogan “Ecology is permanent economy. Sunderlal Bahuguna helped bring the movement to prominence through about 5,000-kilometer trans-Himalaya march undertaken from 1981 to 1983, travelling from village to village, gathering support for the movement. He had an appointment with the then Indian Prime Minister Indira Gandhi and that meeting is credited with resulting in Ms. Gandhi’s subsequent 15-year ban on felling of green trees in 1980. He was also closely associated with Gaura Devi, one of the pioneers of the movement (Wikipedia, Sunderlal Bahuguna).

Anti Tehri Dam Protests

He stood against the anti-Tehri Dam protests for decades, he used the Satyagraha methods, and repeatedly went on hunger strikes at the banks of Bhagirathi as a mark of his protest. In 1995, he called off a 45-day-long fast following an assurance from the then Prime Minister P.V. Narasimha Rao of the appointment of a review committee on the ecological impacts of the dam, thereafter he went on another long fast another fast which lasted for 74 days at Gandhi Samadhi, Raj Ghat, during the tenure of Prime Minister, H.D. Deve Gowda, he gave personal undertaking of project review (Wikipedia, Sunderlal Bahuguna).

Eventually, the dam reservoir started filling up in 2004, and on 31 July 2004 he was finally evacuated to a new accommodation at Koti, a little hillock, along the Bhagirathi where he lives today, continues his environment work.

Sunderlal Bahuguna has been a dedicated to the conservation of the Himalayan ecology and its people, working for temperance, the plight of the hill people (especially working women). He has also struggled to defend India’s rivers.

Narmada Bachao Andolan

Narmada Bachao Andolan is one of the most powerful mass movements, which started in 1985 against the construction of huge dam on the Narmada River. Narmada is the India's largest west flowing river, which supports a large number of people with unique culture and tradition ranging from the indigenous (tribal) people inhabited in the jungles here to the large number of village folk. The proposed Sardar Sarovar Dam and Narmada Sagar will displace more than 250,000 people. The struggle is over the resettlement or the rehabilitation of these people. The two proposals are already under construction, supported by US\$550 million loan by the World Bank. There are plans to build over 3000 big and small dams along the river.

It is multi crore project that will generate a big revenue for the government. The supporters are of the view that it will produce 1450 MW of electricity and pure drinking water to 40 million people covering thousand of villages and towns. Some of the dams have been already been completed such as Tawa and Bargi Dams. But it is environmentally not suitable say this hydro project will devastate human lives and bio diversity by destroying thousands of acres of forests and agricultural land. On the other hand it will deprive thousands of people of their livelihood. They believe that the water and energy could be provided to the people through alternative technological means that would be ecologically beneficial.

Led by one of the well-know environmentalists Medha Patkar, it has now been turned into the International protest, gaining support from NGO'S all around the globe. Protestors are agitating the issue through the mass media, hunger strikes, massive marches and rallies and the through the on screen of several documentary films. Protests have been protesting peacefully, but peaceful they been harassed, arrested and beaten up by the police several times. The Narmada Bachao Andolan has been pressurizing the World Bank to withdraw its loan from the project through media.

Appiko Movement in Karnataka

Vandana Shiva, has pointed out that the people's struggle against commercial forest policy has been significant in the region of Uttara Kannada. The destruction of tropical natural forests has caused irreversible changes in the ecosystem of the forests. The desiccation of mixed species denied people's access to biomass for fodder, fertilizer and other resource. 'The clear felling of natural forests has led to severe soil erosion and drying up of perennial water resources. Moved by the destruction of essential ecological processes, the youth of Salkani village in Sirsi launched a Chipko movement, which was locally known as "Appiko Chaluvali". They embraced the trees which were to be felled by contractors of the forest department' (quoting article shared by Puja Mondal, Appiko Movement in India: useful notes).

The protest within the forest continued for 38 days and finally the felling orders were withdrawn. The success of this agitation spread to other places and the movement has now been launched in eight areas covering the entire Sirsi forest division in Uttara Kannada and Shimoga districts.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Give an introductory note to the human intervention and civilization activities in changing the natural land-scope through the centuries.

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2. How did the Bishnoi Philosophy conserve the natural resource of the Thar Desert?

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3. How did British Colonialism interface with the livelihood of the tribal and forest communities?

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4. Give a brief outline of the National Forest Policy of 1952 and the National Forest Policy of 1988 with reference to Van Mahotsava in India.

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9.9 LET US SUM UP

In this unit we have studied about how Human civilization and human intervention have shaped the contours of natural landscape and natural environment from the Paleolithic age to the modern times the Bishnoi movement in Medieval Western Rajasthan is one of the earliest environmental movements recorded in Indian history. Colonial India: witnessed forest policies, tribal land displacement and deprivation of livelihood of Adivasi. Routine revolts by tribes of forests were a feature of British India. Forest Act of 1865, 1878, 1972 and 1935 tightened colonial control over forests, its resources and its routes. Van Mahotsava was inaugurated under the national forest policy of 1952 to increase forest coverage. This policy was replaced by the National Forest Policies of 1988. Environmental Movements in contemporary India: Chipko movement, Narmada Bachao, Andolan and Appiko Independent India witnessed environmental movements to protest against destruction of natural resources

and for the conservation of nature environmental leaders like Sundar Lal Bahuguna, Vandana Shiva, and Medha Patkar, are reckoning names in today's India.

9.10 KEY WORDS

Environmental movement: It is a term used to build up awareness and a growing organized movement to protect natural environmental.

Conservation of environmental resources: It is a term used to describe preservation and generation of natural resources.

9.11 FURTHER READINGS

Mahesh Rangarajan & K. Sivaramakrisnan (Ed.) India's Environmental History: Colonialism, Modernity, and the Nation 2012, 2013 Ranikhet.

Madhav Gadgil & Ramchander Guha (Ed.) This Fissured Land: And Ecological History of India, 1992 University of California.

Ramchander Guha (Ed.) The Unquiet woods: Ecological change and Peasant resistance in the Himalaya, 2000 University of California press.

Madhav Gadgil & Ramchander Guha (Ed.) Ecology & Equity: The use and Abuse of Nature in Contemporary India, 2013 New Delhi.

Nandini Sinha Kapur (Ed.) Tribes, Agro-tastoralists and Environment in Western India (7- 28th Centaury), 2008 New Delhi.

9.12 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Your answer should include the following points:
 - Human civilization has shaped the contours of natural environmental throughout the centuries.
 - The first four river valley civilizations grew on river banks as urbanized societies exploited water resources and alluvium soil.
 - The Mauryan Empire intruded into forests for wood, elephants, mining wealth and control of strategic routes.
2. Your answer should include the following points:
 - The Bishnoi philosophy propagated conservation of natural Vegetation, and water resources and agricultural practices in the arid climate of Western Rajasthan.
 - The Bishnois Condemned felling of trees and fiercely fought for the protection of Khejri trees.
 - Their teachings for the conservation of nature were equated to Gurudharma (duties for the master, Guru Jambhoji, the founder of the Bishnoi Monument).

3. Your answer should include the following points:

- 1878 Forest Act made administrative categories of forests for the total control of forest resources and routes intervening into the additional livelihood of foresters and tribal's.
- Reserved forests were commercially valuable and subject to exploitation.
- In the Singbhum District of Bihar, large-scale encroachment by the forest Department displaced the HO tribe from their villages in the late nineteenth Century in an attempt to demarcate and preserved forest.

The forest Acts of 1865, 1972 and 1935 were attempts at total control of the forests by the colonial state.

4. Your answer should include the following points:

- The Govt. of India inaugurated Van Mahotsava under the National Forest Policy of 1952.
- Central Board of Forestry of raise area under forest to 100 million hectares. This policy was evaluated by the National Commission on Agriculture.
- The National Forest Act of 1952 was replaced by the National Forest Policy of 1988. The Forest Policy of 1988 was concerned with protection, conservation and development of forests.

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